

CyberAI Cup 2026: Final Architectures, Experimental Logs, and Critical Findings for Packaging Difference Mining, Compliance Auditing, and Flow Classification

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Abstract. The CyberAI Cup 2026 comprised three independent tasks: packaging difference mining, document-grounded compliance auditing, and real-time communication (RTC) flow classification. This paper describes our solution to each. For packaging difference mining, we use a Siamese ConvNeXt-Tiny CenterNet that decodes at stride 2 to resolve marks as small as 8×8 px, trained with fold-scoped synthetic small-mark pairs and combined with 8-view test-time augmentation, Weighted Boxes Fusion and a cross-fold ResNet-18 patch verifier. It achieves **0.9577** F1, and a recall-ceiling analysis shows that its out-of-fold candidate set supports up to 0.9801. For compliance auditing, we map each of 41 checking points to its governing sections of the Export Control Rules, condensing a 184-section policy into 18 sections, and audit all nine evidence documents of an establishment in a single full-context Gemini 3.5 Flash call. This reaches **0.7818** accuracy, and full-context reasoning outperformed hybrid retrieval (BM25, dense retrieval and ColBERT reranking) at lower token cost. For RTC flow classification from only five packets, we blend gradient-boosted trees with the tabular foundation model TabICL and add a parameter-free decision rule derived from a separability analysis of audio-only Zoom windows, reaching **0.8746** accuracy. Our solutions achieved an overall score of 0.8714, the mean of the three task scores, ranking first of 16 teams. We also report the evaluation protocol behind every estimate, together with the negative results and implementation bugs that shaped each design.

1 Introduction

The CyberAI Cup 2026 posed three unrelated machine-learning problems: finding small content differences between packaging designs and photographs of the printed product (Task 1), deciding statutory compliance from several heterogeneous documents with an LLM (Task 2), and classifying real-time communication flows from five-packet traces (Task 3). Table 1 gives our official results.

The approach was the same in every track: build a controlled baseline, locate where it fails before changing the model, and measure what the competition metric does not

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show (recall ceilings in Task 1, per-class and per-element accuracy in Task 2, and an upper bound set by one inseparable class pair in Task 3). Several conclusions changed once we examined the complete pipeline, most visibly in Task 1, where two fusion bugs made test-time augmentation and ensembling look worse than no fusion.

Our contributions are: (i) three complete systems with official results; (ii) for each task, a record of which estimate came from which protocol (out-of-fold, nested, development set, official); (iii) negative results and implementation bugs, including the effect of each on earlier conclusions; and (iv) the limitations that remain, with the experiments we did not run.

Sections 2–4 describe, for each task, the problem, the submitted system, the development path that led to it, and the results. Section 5 draws the lessons common to the three tracks. Appendices A–C contain the full experimental logs, extended tables and per-task limitations; Exp. *An*, *Bn* and *Cn* denote experiment *n* of Task 1, 2 and 3.

Table 1. Official test results (team 1665, rank 1 of 16, overall 0.871353, the mean of the three accuracy values) next to our own estimates.

Task	Acc.	Prec.	Rec.	F_β	Our estimate
1	0.9577	0.9656	0.9499	0.9577	pooled OOF F1 0.9465; nested 0.9435
2	0.7818	0.7758	0.7818	0.7783	dev accuracy 0.861 (10 cases)
3	0.8746	0.9211	0.8746	0.883	CV accuracy 0.8389; macro-F1 0.8318

2 Task 1: Packaging Material Difference Mining

2.1 Problem and Data

Given a clean design template and a photograph of the corresponding printed packaging, the system must place a box around every genuine content difference between the two. Because photographs are re-scanned, re-lit and re-toned versions of the template, the difficulty is not detecting *some* difference but separating genuine content edits from scan noise, blur and shadow. Predictions are scored by a single *global* (pooled, not per-image) F1 at IoU ≥ 0.5 , with a stated target of 0.97. The organisers report accuracy, precision, recall and F_β ; for our submission accuracy and F_β coincide (0.9577), so we use F1 throughout.

The training set has 200 pairs with 1,407 boxes and the test set 100 pairs (Table 4 in Appendix A). Two properties dominate the design space: boxes are small (median 22×22 px, with 83 boxes exactly 8×8 px), and the changes are asymmetric (842 additions, 558 modifications and no pure deletions).

2.2 Submitted System

The pipeline (Fig. 1) has five stages.

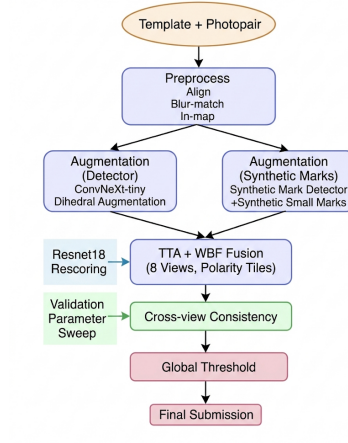


Fig. 1. Task 1 pipeline: preprocessing, a ten-detector Siamese CenterNet ensemble, 8-view TTA fused by WBF, five cross-fold ResNet-18 verifiers and the final decode (threshold 0.43, swept on out-of-fold predictions).

Stage 0: preprocessing. A phase-correlation alignment check, with translation correction if needed; blur matching of the template to the photo by a σ -grid search on Laplacian variance; and a shadow- and tone-invariant *ink map*, $\text{GaussianBlur}(\text{gray}, \sigma=31) - \text{gray}$.

Stage 1: detector. A Siamese ConvNeXt-Tiny [2] encoder with shared weights over 4-channel (BGR + ink) streams, fused at every scale by $\text{conv}(\text{concat}(a, b, |a - b|))$ and decoded through a U-Net [4] path to **output stride 2** rather than the usual 4, to resolve the 8×8 px boxes (stride 4 was not ablated). Four CenterNet-style [1] heads predict a centre heatmap, width/height, a sub-pixel offset and an auxiliary change mask. Two training recipes $\times$ five folds give ten detectors: `_aug` uses the real training pairs with dihedral augmentation, and `_augsyn` adds fold-scoped synthetic small-mark pairs (Exp. A5).

Stage 2: TTA fusion. Each detector runs on 8 dihedral views, fused with Weighted Boxes Fusion [5]; the detectors' fused outputs are then pooled.

Stage 3: verifier. A ResNet-18 [3] on 96×96 crops with an 8-channel stem produces a real-difference probability, blended with the detector score, and a box-delta refinement. Five verifiers are trained strictly cross-fold.

Stage 4: final decode. A polarity filter (no pure-deletion class exists in the data), ink-map snapping with a measured margin, and one global score threshold (0.43), swept only on out-of-fold predictions.

Submission. The submission contains 668 boxes over the 100 test images (6.68 per image, against a training average of 7.04), from all ten detectors with TTA and all five verifiers averaged. The out-of-fold evaluation used a smaller system (per fold, one detector of each recipe and one verifier), so the official score is the only estimate that covers the submitted system.

Leakage discipline. (1) Synthetic pairs are fold-scoped: each fold discards the synthetic pairs derived from its own validation pages (836–892 of 6,400 per fold). (2) Unlabelled

test templates, which carry no labels, are used as a synthesis source; no test photographs are used for threshold selection. (3) The verifier for fold k trains only on candidates from folds $\neq k$, produced by detectors that never saw those pages. (4) The global threshold is swept on out-of-fold predictions only, never on the reported fold or the test set.

2.3 Development Path

From the first Siamese CenterNet (fold-0 F1 0.9049) to the pooled detector-plus-verifier system (0.9465), four observations decided the design; Table 5 in Appendix A traces every step.

Precision is the hard part. A classical difference baseline reaches recall 0.572 but F1 0.0293, because every glyph edge fires in the blurrier photo (Exp. A1). Blur matching and the ink map of Stage 0 follow from this.

Model and decode are entangled. The same `_aug` weights scored 0.9049 at the original decode and 0.9257 with a lower threshold, a larger box cap and 8-view TTA (Exp. A6), so earlier “no gain” verdicts had compared decoders as much as models.

The recall ceiling, not F1, chose the ensemble. The fraction of ground-truth boxes present anywhere in the candidate list bounds every rescoring idea. `_augsyn` has worse standalone F1 (0.896 against 0.926) but raises the ceiling from 0.9440 to 0.9590, and on boxes under 12 px from 0.891 to 0.922 (Exps. A5, A8). The two-family ensemble adds only 0.001 F1 but lifts the ceiling to 0.963, leaving the verifier to do the rescoring (Exp. A9). The verifier first gave +0.0009 because it was trained on filtered, capped candidates with almost no real negatives (Exp. A10); retrained on about 11,000 raw candidates per fold at 8:1 negatives to positives, it improved every fold, by +0.006 to +0.018 (Exp. A11).

Silent bugs made working ideas look like failures (Exps. A16–A18). Fused WBF confidence was computed as $\text{sum}(\text{scores})/\text{min}(\text{members}+1, 3)$, which clipped well-supported boxes to 1.0 while single-view boxes kept their raw score, so 8-view TTA scored F1 0.775, worse than no TTA. The ensemble also double-counted its sources, collapsing recall to 0.056, and a verifier loader that re-decoded full pages took about 60 minutes per epoch with the GPU idle. All numbers from Exp. A6 onwards were produced after the fixes; the size-relative box loss (A3) and EMA weights (A4) predate them, showed no effect and were not repeated.

2.4 Results

The organisers scored the submission at accuracy 0.9577, precision 0.9656, recall 0.9499 and F_β 0.9577 (Table 2); they released only these rounded metrics, not confusion counts. With 668 predictions, back-solving from the rounded precision and recall gives $TP \approx 645$, $FP \approx 23$ and $FN \approx 34$ (implying about 679 ground-truth boxes and $F_1 = 1290/1347$). These are approximate reconstructions, accurate only up to the rounding of the official figures, not reported counts. That is roughly 57 errors, where $F_1 = 0.97$ would have needed about 40 total errors (at most 39 to exceed 0.97, since $2 \cdot 654/1347 = 0.9711$). The larger submitted ensemble is consistent with the upside we wrote down before the results were released, but we cannot separate it from test-set difficulty. Per-fold F1 was 0.943, 0.955, 0.935, 0.961 and 0.968. Threshold-selection optimism, measured by nested

Table 2. Task 1 results. [†]Approximate reconstruction from the official precision, recall and the 668 predictions; the organisers did not release confusion counts.

Configuration	F1	Prec.	Rec.	TP	FP	FN
Official test (accuracy = F1)	0.9577	0.9656	0.9499	$\approx 645^\dagger$	$\approx 23^\dagger$	$\approx 34^\dagger$
Nested out-of-fold (threshold-bias-corrected)	0.9435	—	—	—	—	—
Pooled out-of-fold, detector + verifier (A12)	0.9465	0.963	0.930	1,309	50	98
Pooled out-of-fold, detector only (A13)	0.9385	—	—	—	—	—
Fold-0 starting point (A2)	0.9049	0.922	0.888	238	20	30
Perfect-rescorer ceiling (A15)	0.9801	1.000	0.961	1,352	0	55
Competition target: 0.97 (not reached; practically out of reach with these proposals)						

cross-validation, is 0.0030 (Exp. A13). Checkpoint selection inflates the undertrained _augsyn models by +0.034 on average (Exp. A14), but the official score still exceeded the raw out-of-fold F1.

Qualitative results. Fig. 2 shows example out-of-fold predictions: correct detections, false positives, boxes that no model proposes (the 55 boxes of Exp. A15), and candidates whose score the verifier changed (Exp. A11).

Reachability of 0.97. The final candidate set contains 1,352 of the 1,407 boxes, so even a perfect rescorer caps at F1 0.9801 (Exp. A15). The target lies below this ceiling, but keeping all 1,352 matching proposals would still allow at most about 28 false positives, against 50 out of fold, and the achieved 0.9465 is 0.0336 below the ceiling. Further gains need better proposals for the 55 boxes no model finds, not better verifiers.

Limitations. Stride 2, the ink map, Siamese fusion, the polarity filter, ink-map snapping and box-delta refinement were design choices that were not ablated individually. Fold-0 comparisons rest on 268 boxes with no multi-seed noise estimate. Exps. A3 and A4 were not repeated under the corrected fusion.

3 Task 2: FRECA, Compliance Auditing with Grounded LLM Reasoning

3.1 Problem and Data

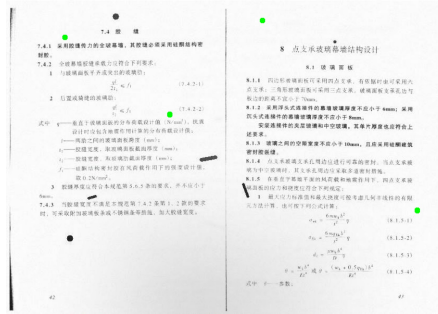
Given nine heterogeneous evidence files per farm export establishment (Registered Establishment, RE), the system must audit compliance against 41 checking points (CPs) drawn from the *Export Control (Plants and Plant Products) Rules 2021*, emitting 1 (compliant), 0 (non-compliant) or N/A per CP. Three hard constraints govern the design: no hard-coded compliance logic anywhere in the pipeline, LLM-API-only reasoning, and reproducibility.

The corpus has 99 raw folders that resolve to 100 logical cases, each with 7 .docx and 2 .xlsx tracks, and a 132-page policy document with 184 sections (Table 6 in Appendix B). No labels were provided; we hand-labelled a 10-case development set for internal measurement only, and its labels were not independently verified. A median case is about 8.2K words (22–26K tokens), so a whole case fits in one context window.

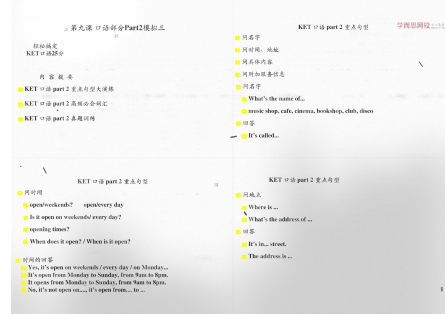
(a) Correct detections



(b) False positives



(c) Missed boxes (never proposed)



(d) Verifier corrections

Fig. 2. Task 1 qualitative results on out-of-fold predictions, with ground-truth and predicted boxes overlaid on each pair.

Exploratory analysis (Exp. B1) found that one folder holds two complete establishments under the same RE number (the “missing” 100th case), two folders lack the registration form, and some apparent signals are generation noise.

3.2 Submitted System

The pipeline (Fig. 3) has four phases.

Phase 1: Open Knowledge Format (OKF), offline, once. The policy PDF is chunked into 184 sections by a deterministic regex on section headers. A one-time LLM pass reads the statute text and maps each of the 41 CPs to its governing section(s), producing a YAML-cited bundle of 18 unique sections (deduplicated from 64 citation instances). It is stored with the system and can be re-derived on demand.

Phase 2: deterministic evidence ETL, per case. A canonical logical-case manifest resolves the 99 folders into 100 cases by splitting the doubled folder; an ingestion step flattens .docx/.xlsx files into track-tagged markdown, preserving table structure, tolerating missing tracks and sorting lexicographically for reproducibility. This layer contains no compliance logic.

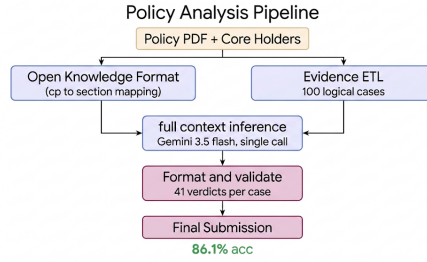


Fig. 3. Task 2 pipeline: an offline Open Knowledge Format (CP-to-section grounding) and a per-case deterministic evidence ETL feed a single full-context call to Gemini 3.5 Flash, followed by formatting and validation into the submission matrix.

Phase 3: full-context inference, per case. One prompt per case (~44K tokens) holds the complete evidence, the OKF bundle (CP texts and governing sections) and a persona prompt enforcing identity anchoring, boilerplate-versus-signal discipline and N/A discipline. Model: Gemini 3.5 Flash [14], temperature 0, reasoning enabled; temperature 0 does not guarantee identical outputs across runs. Output: structured JSON with 41 verdicts, reasoning and confidence.

Phase 4: formatting and validation. Verdicts are enum-validated into the submission matrix (RE Number + CP1...CP41).

The ETL performs pure extraction and the CP-to-section mapping is LLM-derived from the statute, so neither contains compliance rules. The persona prompt, however, carries a CP38-motivated instruction added after Exp. B9. We distinguish encoding a checking point’s *requirement* (what the governing text obliges) from hard-coding the *expected answer* (instructing which verdict to emit for a specific evidence string); the former is legitimate grounding, the latter is non-compliant. The added wording is general in form — “where a register row has both a fixed target and a case-specific status/verification note, the status note is the signal, not the target” — but its examples name the concrete strings “2-year retention verified” and “migration in progress” / “pending” that separate compliant from non-compliant rows in this dataset. It is therefore a general interpretation rule with CP38-answer-specific examples, so the no-hard-coding constraint holds only partially. The intervention was developed and scored on the same ten dev cases and was never evaluated on unseen cases, so we do not claim the constraint is fully satisfied.

Production run (Exp. B14). With a 240-second per-call timeout and per-case checkpointing, all 100 cases completed with no blanks and no parse failures, using 4,493,869 tokens (~44.9K per case); the run was resumed four times after key changes and credit or rate-limit stops without losing progress. The output holds 3,118 1, 982 0 (23.95%, against 17.6% in the dev gold) and no N/A. The 100th case, the second establishment in the doubled folder, was submitted under a provisional ID.

3.3 Development Path

Model choice. With identical prompts, grounding and evidence (Exp. B4; Table 7 in Appendix B), Gemini 3.5 Flash reached 86.1% against 76.6% for Nemotron Ultra and 81.5% for DeepSeek V4 Pro. Sonnet 5 scored highest (91.0%) but was withdrawn because the submission had to be reproducible through the team’s own API key (an access constraint, not a quality judgement); Gemini 3.5 Flash was the best model that met it.

Full context beat retrieval. A hybrid RAG branch [11] (BM25 [10] and bge-small dense retrieval [13], weighted fusion, ColBERT [12] rerank) scored 74.9% at 610K tokens against 86.1% at 442K for full context (Exp. B6). When a case fits in one context window, retrieval only adds the risk of missing a decisive sentence: the “maintained in Mandarin” sentence was retrieved only ~56% of the time.

Fewer calls. The call architecture went from per-element, per-persona calls (13 per case, megabyte-scale prompts) to a 3-persona ensemble (3 calls, ~215K tokens, with a hard-coded majority threshold) to the single full-context call (~44K tokens) (Exp. B8). On one case a Haiku persona ensemble scored 70.7% against 92.7% for a single Sonnet pass at lower token cost (Exp. B5); this confounds model tier with ensembling, and we did not test a strong-model ensemble.

Boilerplate versus signal. CP38 (record retention) was being decided from a date-range string that is byte-identical across compliant and non-compliant cases. Telling the model that the per-case status note is the signal took CP38 from 4/10 to 10/10 on the in-sample dev set. This was measured on the Sonnet 5 development run, *not* on the submitted Gemini 3.5 Flash model, so it is a prompt-development observation rather than a result of the submitted system; the same class of fix is still needed for CP37 (Exp. B10).

Error analysis on the dev set (Exps. B10–B13): the model never emits N/A (all 8 gold-N/A cells were predicted 1), Element-4 (traceability) is weakest at 77.7%, and CP39’s gold is internally inconsistent.

3.4 Results

The organisers scored the submission at accuracy 0.7818, precision 0.7758, recall 0.7818 and F_β 0.7783. Recall equal to accuracy indicates support-weighted averaging. The 0.861 figure is *in-sample development accuracy* on a 10-case set that we self-labelled and then used to tune the prompts (including the CP38 wording); it is not an estimate of generalization, and we do not report it as such. The official accuracy is 7.9 points below that in-sample number and below the worst dev case (dev cases range from 80.5% to 95.1%; Exp. B11), so the gap is not sampling noise within the dev set. However, because no independent labelled evaluation set exists, the cause of the official-score gap *cannot be established from the available evidence*; the following are hypotheses only: the dev labels are self-assigned and unverified; the CP38 prompt was tuned on the dev set; CP39’s gold is inconsistent; and $n = 10$. Over-flagging is a further hypothesis, since the production run predicts 0 more often than the dev gold and class-0 precision on the dev set was only 0.616.

Limitations. The ten dev cases are self-labelled and were used to tune prompts, so dev accuracy is not held-out accuracy; the CP37–39 analysis used the Sonnet run rather than the submitted model; the OKF was not ablated, and only 12 of 41 CPs were mapped

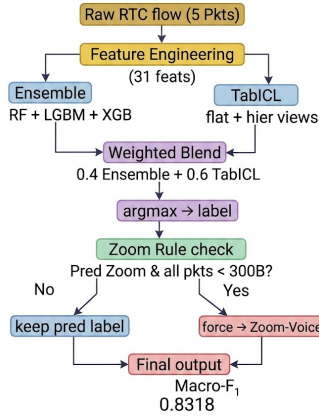


Fig. 4. Task 3 pipeline: 31 columns feed the tree ensemble (Branch A) and 241 columns feed TabICL in flat and hierarchical views (Branch B); the probabilities are blended 0.4/0.6, and a parameter-free rule sends audio-only Zoom windows to Zoom_voice.

identically by two mapping models; ensembling was not tested with a strong base model; and the exact model snapshot is not recorded (Appendix B.12).

4 Task 3: Real-Time Communication Flow Classification

4.1 Problem and Data

Given only the first five packets of a real-time communication (RTC) flow (packet lengths and relative timestamps), the system must predict which of five applications (Discord, Google Meet, Messenger, WhatsApp, Zoom) produced the flow and whether it is a voice or a video call, giving ten classes. The organisers report accuracy, precision, recall and F_β ; for our submission recall equals accuracy, which points to support-weighted averaging. During development we optimised macro-F1, which weights the small classes (40 Google Meet voice flows) as much as the large ones (256 Discord video flows).

There are 1,285 training and 327 test flows (Table 10 in Appendix C), and train and test are exchangeable (adversarial-validation AUC 0.529; Exp. C1). Three properties dominate the design space: no call identifier is released although flows from the same call are near-duplicates (an estimated ≈ 400 source calls); more than half of all flows (56.5%) have a window in which every packet is below 300 bytes, i.e. only audio-sized packets whatever the call mode; and the dataset is small.

4.2 Submitted System

The pipeline (Fig. 4) has five stages and two feature representations.

Stage 0: features. Branch A uses 31 compact columns: five lengths, four timestamps, four length differences, four inter-arrival gaps, the lengths in rank order, argmax/argmin position, range, mean, standard deviation, median and total span, plus the logarithm of the fourth packet length and the position of the largest packet-to-packet size increase. Branch B uses 241 deterministic columns (raw and log values, inter-arrival and length statistics, sequence shape, protocol bands and residues, burst structure, rate and spectral terms). No feature depends on labels.

Stage 1: Branch A, tree ensemble. RandomForest [6] (600 trees, balanced-subsample weights), LightGBM [8] (400 trees, depth 4, 15 leaves, learning rate 0.05, balanced) and XGBoost [7] (400 trees, depth 4, learning rate 0.05, subsample and column sample 0.8). Probabilities are averaged with equal weight, softened by a temperature $T = 2$ ($\tilde{p}_k \propto \bar{p}_k^{1/T}$), divided by the empirical class prior $\hat{\pi}_k$ and renormalised. All estimators run single-threaded (Exp. C9).

Stage 2: Branch B, TabICL. TabICL [9] with 16 internal estimators and no tuning, in a flat ten-class view and a hierarchical view (an application head and per-application voice/video heads); the two probability vectors are averaged.

Stage 3: blend. $p = 0.4 p_A + 0.6 p_B$, then argmax. The weight is the median of ten weights, each chosen on the training side of one grouped fold (Exp. C23).

Stage 4: Zoom rule. If the blended label is either Zoom class and all five packet lengths are below 300 bytes, the label is set to Zoom_voice. The rule has no fitted parameter (Exps. C18–C19).

Validation protocol. Every Branch B selection decision is scored on grouped folds (StratifiedGroupKFold, 5 splits $\times$ 2 repeats) over a pseudo-call key, the packet-length tuple rounded to 8 bytes joined with the rounded log-duration, which yields 1,107 pseudo-groups. Negative controls pass (shuffled labels 0.1549, prior-only 0.1992; Exp. C2). Only nested values of fitted quantities (blend weight, offsets, biases) are used for choosing. Branch A, developed earlier under stratified folds, is the one exception. Unlabelled test flows are never used to fit a label-dependent quantity.

Submission (Exp. C25). Branch A was refit on all 1,285 rows *without* the per-class offsets and specialist of Exp. C7, so that the blend receives corrected probabilities rather than outputs shifted toward recall; Branch B test probabilities come from a full-data refit. The rule routed 43 of the 327 test flows to Zoom_voice, in line with the training proportion.

4.3 Development Path

Model search. 2,377 grid points over eleven model families, followed by Optuna [15], put boosted trees ahead; distance, kernel, linear and neural models trailed by 9–11 points (Exp. C3). TabICL with no tuning beat the best searched point (LightGBM, 0.8249) at every feature width, though the paired difference is not significant (66 rows won against 49, $p = 0.135$; Exp. C10). Feature pruning, label-aware features, pretraining on a public corpus [21], probe features, a tuple lookup and call-level aggregation did not help (Exps. C4, C12, C13, C15).

One pair bounds the score. The standing tree configuration made 237 errors, 175 of them in audio-only windows and 68 between Zoom_voice and Zoom_video alone

Table 3. Task 3 results. Cross-validation rows are pooled grouped out-of-fold predictions (each of the 1,285 rows scored twice); the first two rows are single-model references (Exps. C3, C10).

Configuration	Accuracy	Macro-F1	Fold mean
Official test (weighted metrics, $n = 327$)	0.8746	—	—
Best searched point, LightGBM (C3)	0.8249	0.8069	—
TabICL, default, 241 columns (C10)	0.8389	0.8232	—
Branch A, corrected probabilities (stratified folds)	0.7961	0.7900	—
Branch B, flat+hier average	0.8358	0.8201	0.8175
Branch B + Zoom rule	0.8342	0.8292	0.8268
Blend, nested weight, no rule	0.8389	0.8319	0.8286
Blend, nested weight, + Zoom rule (submitted)	—	0.8345	0.8318
Blend, $w = 0.40$, + Zoom rule ^a	0.8389	0.8366	—

^a w located on the same rows (Exp. C23); optimistic.

(Exp. C17). Inside audio-only windows the Zoom pair is not separable: the subset is exactly balanced (80 voice, 80 video) and the grouped AUC of video against voice is 0.551, against 0.83–0.88 for the other applications (Exp. C18). Even with every other flow correct, accuracy is capped at 0.9377 and macro-F1 at 0.9364.

A derived rule, not a fitted one. On an evenly split, inseparable subset every assignment has the same accuracy but not the same macro-F1: sending all 160 flows to Zoom_voice gives F1 0.667 (voice) and 0.697 (video), against 0 and 0.811 for Zoom_video (Exp. C19). The rule improved macro-F1 for all six model families in every view (Exp. C20), whereas fitted per-class biases gained in sample and lost nested (Exps. C14, C21). It is neutral for accuracy (0.8233 $\rightarrow$ 0.8241).

Two branches. Branch A’s complementary strength is Zoom_voice recall (0.712 against 0.438 for Branch B; Exp. C22). Nested selection picked blend weights from 0.25 to 0.55 across folds, median 0.40 (Exp. C23). The blend improves on Branch B with the rule by +0.0039 to +0.0072 macro-F1, but the bootstrap interval includes zero (Exp. C24).

4.4 Results

The organisers scored the submission at accuracy 0.8746, precision 0.9211, recall 0.8746 and F_β 0.883 on the 327 test flows. Because the official metrics are support-weighted, F_β is not comparable with our macro-F1 estimates in Table 3. The test accuracy is 3.6 points above the cross-validated accuracy of the blend (0.8389); with $n = 327$ the standard error of an accuracy near 0.87 is about 1.9 points, so the difference is about two standard errors. Possible contributors are training on all 1,285 flows rather than four fifths and the weighted metric favouring large classes; we did not test either.

Limitations. Branch A’s out-of-fold probabilities come from stratified rather than grouped folds, so the 0.40 weight on Branch A is, if anything, too high and the nested estimate of 0.8318 is likely somewhat optimistic. The Branch B variant, the 16-estimator setting and the blend were chosen among near-equal candidates on the same 1,285 rows. The Zoom rule is a property of this dataset’s Zoom audio-only windows and is not claimed to transfer (Appendix C.11).

5 Cross-Task Discussion

Development estimates and official scores disagreed, in both directions. Task 1’s pooled out-of-fold F1 of 0.9465 became 0.9577 officially and Task 3’s cross-validated accuracy of 0.8389 became 0.8746, while Task 2’s dev accuracy of 0.861 became 0.7818, outside the spread of its own dev cases. In Tasks 2 and 3 official recall equals accuracy, so the metric is support-weighted, and our macro-averaged concerns (Task 2) and macro-F1 objective (Task 3’s rule, neutral on accuracy) did not transfer.

Secondary diagnostics show where the headline metric is bounded. The Task 1 recall ceiling (0.9801) lies 0.0336 above the out-of-fold F1 (Exps. A8, A15); Task 2’s per-class and per-element breakdowns exposed the missing N/A class and the weak traceability element (Exps. B12, B13); and one inseparable class pair caps Task 3’s macro-F1 at 0.9364 (Exp. C18).

Negative results need a bug hypothesis first. Task 1’s verifier looked like a dead end (+0.0009, Exp. A10) until it was retrained on raw candidates and gained on all five folds (Exp. A11), and fusion bugs had made TTA and ensembling look harmful (Exps. A16, A17). Data structure is likewise an engineering problem and a leakage risk: fold-scoped synthesis (Task 1), a logical-case manifest (Task 2, Exp. B1) and pseudo-call groups (Task 3, 1,107 groups against an estimated 400 calls, so grouped cross-validation may still leak). The practical lesson is to state next to every number which protocol produced it, and to treat the organisers’ held-out scores as the only reliable estimate.

6 Conclusion

Officially, Task 1 scored **0.9577** (out-of-fold 0.9465; candidate-set ceiling 0.9801), Task 2 scored **0.7818** with a single full-context LLM call (development estimate 0.861), and Task 3 scored **0.8746** accuracy with gradient-boosted trees, TabICL and a deterministic rule (pooled grouped out-of-fold accuracy 0.8389). The common lessons were to separate model quality from decoding and scoring effects, to measure what the headline metric hides, and to suspect the experiment before the idea when a result is unexpectedly negative.

Compute Task 1 detectors and verifiers were trained on A100 GPUs on Modal.

Use of large language models Gemini 3.5 Flash is the reasoning component of the Task 2 system. Claude Sonnet 5 was used in the Task 2 development bake-off and per-checking-point analysis (Exps. B4, B9, B10) but not in the submitted system. Claude was used to restructure and copy-edit this manuscript; the authors checked all content.

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A Task 1: Dataset, Full Experimental Log and Bugs

A.1 Dataset and Data Usage

Table 4 gives the full dataset statistics. The change-polarity labels (842 additions, 558 modifications, zero pure deletions) sum to 1,400 of the 1,407 boxes. The unlabelled test templates were used as a source for synthetic training pairs (Experiment 5). No test-set labels exist to us, and no test photographs were used for threshold selection.

Table 4. Task 1 dataset statistics.

Quantity	Value
Train pairs / boxes	200 / 1,407
Test pairs	100
Dataset size	2.5 GB
Boxes per image	3–11, median 7
Median box size	22×22 px
Boxes under 16×16 px (both sides)	419
Boxes exactly 8×8 px	83
Change polarity	842 additions, 558 modifications, 0 deletions (7 boxes unlabelled)
Alignment	no warp needed for 284/300 pairs, p95 residual ≤ 0.26 px
Photo degradation	2–6 $\times$ lower Laplacian variance than template; $\sim 18\%$ heavy shadow

A.2 Development Path Summary

Table 5 summarises the steps from the first baseline to the pooled five-fold result (Section 2.3).

Table 5. Task 1 development path. Fold-0 out-of-fold F1 (268 boxes; one box = 0.004 F1) unless marked as pooled over five folds (1,407 boxes).

Step	Exp.	F1	Note
Classical normalised difference	A1	0.0293	recall 0.572, precision 0.015
Siamese CenterNet, single model	A2	0.9049	reference control
_aug weights, old $\rightarrow$ new decode	A6	0.9257	same weights, +0.021
+ _augsyn detector (WBF ensemble)	A9	0.9266	ceiling 0.944 $\rightarrow$ 0.963
+ cross-fold verifier	A11	0.9430	gain on all five folds
Pooled, detector only	A13	0.9385	nested threshold: 0.9385
Pooled, detector + verifier	A12	0.9465	nested threshold: 0.9435

A.3 Organisation of the Log

Nineteen experiments were numbered in sequence; we group them by theme below. Experiments 6 and 7 are the same study and are reported together as Experiment 6 (the number 7 is unused); Experiment 19, an infrastructure failure with no effect on any result, is described in Appendix A.11. Within this appendix, “Experiment n ” refers to Experiment A_n . Unless noted, results are reported on fold 0 out-of-fold validation before the full 5-fold run (Experiment 12). Numbers in Experiments 6 and later were produced after the fusion bug of Experiment 16 was fixed; Experiments 3 and 4 were not repeated (see the limitations in Section 2.4).

A.4 Baseline establishment

Experiment A1: Classical normalized-difference baseline

Hypothesis	A naive pixel-wise difference between template and photo can localize content changes directly, since differences should stand out after alignment.
Method	Compute normalized pixel difference between aligned template and photo, threshold to get blobs, convert blobs to boxes.
Result	Recall 0.572 at precision 0.015. Global F1 = 0.0293.
Inference	Fails catastrophically because every glyph edge fires — the photo is systematically softer/blurrier than the template, so raw differencing cannot distinguish scan/print noise from genuine changes. Precision, not recall, is the hard part of this task. Blur-matching template to photo before differencing is a prerequisite, not an optimization.
Verdict	Rejected as a primary approach; directly informed the design of the Stage 0 ink-map preprocessing.

Experiment A2: Siamese CenterNet, single model, fold 0 (reference baseline)

Hypothesis	A shared-weight ConvNeXt-tiny Siamese encoder over 4-channel (BGR + ink) streams, decoded through a U-Net path to output stride 2 (instead of the standard stride 4), can localize the small (median 22×22 , many 8×8) ground-truth boxes at sub-pixel precision.
Method	Train a Siamese CenterNet with four heads (centre heatmap, w/h, sub-pixel offset, auxiliary change mask) on 768 px tiles, 60% sampled around annotated differences. Decode at <code>score_thr= 0.05</code> , <code>max_boxes= 40</code> , no TTA.
Result	Fold 0 OOF: F1 = 0.9049, Precision = 0.922, Recall = 0.888 (TP 238 / FP 20 / FN 30 on 268 boxes).
Inference	Establishes that stride-2 small-object detection is viable and sets the control number (0.9049) that every subsequent change is measured against.
Verdict	Kept as the baseline architecture; became the reference for all fold-0 controlled experiments.

A.5 Detector-side ablations

Experiment A3: Size-relative width/height loss

Hypothesis	Standard w/h regression loss is size-blind and should under-penalize errors on tiny (8×8) boxes relative to large ones; a size-relative loss should improve small-box localization.
Method	Replace absolute w/h loss with a size-normalized variant; train on fold 0 with identical settings otherwise.
Result	Fold 0 finished at TP 238 / FP 20 / FN 30 — exactly the reference’s counts.
Inference	No measurable effect. One box is worth 0.004 F1 on 268 validation boxes, and we estimate run-to-run noise at about ± 0.02 F1. Identical TP/FP/FN counts to the reference are unusual at that noise level, so they may mean that the flag did not take effect or that the log was copied from the baseline run.
Verdict	Dropped; kept as an option, disabled by default.

Experiment A4: EMA weights (decay 0.999)

Hypothesis	Exponential moving average of weights should stabilize training and yield a better final checkpoint than raw weights.
Method	Track an EMA of model weights during training; compare EMA versus raw weights at each checkpoint on fold 0 validation.
Result	EMA weights were never selected over raw weights until the final two epochs, where they tied.
Inference	No consistent benefit within the training budget used; on a short schedule EMA does not have time to average out enough noise to help.
Verdict	Neutral; dropped as default, kept behind a flag.

Experiment A5: Synthetic small-mark pairs

Hypothesis	The model under-detects sub-12 px boxes because there are too few real examples; synthetically generating small-mark differences on real page layouts should raise coverage of that size band.
Method	Generate synthetic template/photo pairs with invented small-mark edits on real page layouts, fold-scoped to avoid leakage (Section 2.2). Train a model variant (<code>_augsyn</code>) on real + synthetic data.
Result	Standalone F1 is <i>worse</i> than the augmented-only model (0.896 vs. 0.926). But candidate recall ceiling (fraction of ground-truth boxes present anywhere in the candidate list at IoU 0.5) rose from 0.891 (<code>_aug</code> , the matched control without synthetic data) to 0.922 on the sub-12 px band specifically, and the overall ceiling rose from 0.9440 to 0.9590. (Comparing with the original model’s 0.828 would also credit the dihedral augmentation, which <code>_augsyn</code> shares with <code>_aug</code> .)
Inference	This model looks like a failure on F1 alone because it proposes very aggressively (10,238 candidates vs. 421 for the original, though these counts come from different decode settings and are not like-for-like), but it recovers boxes no other model ever proposes.
Verdict	Kept, but only as an ensemble member, never standalone.

A.6 Decode-time optimization and the model/decode entanglement

Experiment A6: Decode settings and the model/decode interaction

Hypothesis	Flips and transposes are free label-preserving augmentation and should improve generalization; if it appears not to help, the measurement method (decode settings) should be suspected before the change itself. A low score threshold, a higher box cap and 8-view TTA should surface more true candidates, at the cost of noise that WBF and the verifier can filter.
Method	Train with dihedral augmentation (<code>_aug</code>); evaluate the <i>same weights</i> first at the old reference decode (<code>score_thr= 0.05</code> , <code>max_boxes= 40</code> , no TTA), then at <code>score_thr= 0.02</code> , <code>max_boxes= 300</code> with 8-view TTA fused by WBF [5]. Sweep threshold, box cap and view count on fold-0 out-of-fold predictions.
Result	0.9049 at the old decode (looked worthless) versus 0.9257 at the new decode: the same weights, +0.021 F1, about five boxes on 268.
Inference	Because the weights are identical, training-seed noise does not enter this comparison; the remaining uncertainty is the validation sample. A paired bootstrap over images would give an interval; we did not run it. Every earlier “no gain” verdict on augmentation had been measured through a lossy decoder, so comparing models at a fixed default decode compares decoders as much as models. This was the most consequential methodological correction in the project.
Verdict	Kept: augmentation is valuable but only visible once decode is fixed; <code>score_thr= 0.02</code> , <code>max_boxes= 300</code> , 8-view TTA with WBF became the standard decode.

A.7 Ensembling and recall-ceiling diagnostics

Experiment A8: Recall ceiling measurement across model variants

Hypothesis	F1 alone cannot reveal whether a model’s failures come from missing candidates (unrecoverable) or from mis-scored candidates (recoverable by rescoring); ceiling analysis will separate these.
Method	For each model/combination, compute the fraction of ground-truth boxes present anywhere in the candidate list at IoU 0.5, regardless of score, on fold 0 with TTA at score 0.02. Compare against the sub-12 px ceiling specifically.
Result	Original: ceiling 0.9403 (sub-12px 0.828), F1 0.9049. <code>_aug</code> : ceiling 0.9440 (0.891), F1 0.9257. <code>_augsyn</code> : ceiling 0.9590 (0.922), F1 0.8956 (worse F1, much better ceiling). <code>_aug+_augsyn</code> combined: ceiling 0.9627 (0.9375), F1 0.9266 (best two-model combo). A third model (<code>_augpix</code>) is reported with ceiling 0.959.
Inference	Nothing downstream can recover a box that was never proposed, so the ceiling bounds every rescoring idea. This measurement redirected the whole plan: the synthetic model’s poor F1 was misleading, and the choice of two

detector families was made before any expensive rescoring experiment was run.

Verdict Kept as the standing diagnostic; drove the choice of exactly two detector families (`_aug` + `_augsyn`) per fold for the ensemble.

Experiment A9: Two-model ensemble (plain-augmented + synthetic-trained)

Hypothesis Combining the `_aug` and `_augsyn` detectors via WBF should capture both models' coverage while the verifier later separates true positives from the extra noise the synthetic model introduces.

Method Fuse candidates from both detectors (each with its own TTA) via Weighted Boxes Fusion on fold 0.

Result F1 $0.9257 \rightarrow 0.9266$ (+0.001), recall +0.011, ceiling $0.944 \rightarrow 0.963$.

Inference A modest direct F1 gain but a large ceiling gain — this sets up the verifier (Experiment 11) to do most of the real work, since the candidates it now needs to rescore contain far more of the true positives.

Verdict Kept; became the standard ensemble configuration (2 detectors $\times$ 5 folds = 10 detectors total).

A.8 Patch verifier

Experiment A10: Patch verifier, first attempt

Hypothesis A ResNet-18 verifier over 96×96 crops around each candidate box, producing a real-difference probability, should sharpen precision by rescoring detector output.

Method Train the verifier on post-processed candidates: capped at 40 per image, already polarity-filtered.

Result +0.0009 F1 — effectively no gain.

Inference The verifier was starved of negatives: because training candidates were already filtered and capped, it almost never saw a genuine false positive to learn from. This looked like a dead end, but the cause was the training-candidate distribution (already filtered and capped at 40 per image), not a flawed idea; Experiment 11 retrains on raw candidates.

Verdict Initially written off; later revived after fixing the underlying data pipeline.

Experiment A11: Patch verifier, retrained on generous raw candidates

Hypothesis If the verifier is trained on the full unfiltered candidate distribution (not the post-processed one), it should see enough real negatives to learn a useful precision boost, and should generalize if trained strictly cross-fold.

Method Retrain the verifier per fold on $\sim 11,000$ raw candidate records per fold at an 8:1 negative:positive ratio, generated only from candidates the verifier's own fold never trained the detector on.

Result Fold-by-fold F1 delta: fold0 +0.016 ($0.9266 \rightarrow 0.9430$), fold1 +0.014, fold2 +0.006, fold3 +0.006, fold4 +0.018. Positive on all five folds independently.

Inference	The verifier is a genuine, consistent improvement once trained on properly distributed data — the earlier null result (Experiment 10) came from training on filtered, capped candidates that contained almost no real negatives, not from a flaw in the approach. The sign is the same on five disjoint folds, which makes a chance effect unlikely, but each fold has about 40 images. The mean of the per-fold deltas (+0.012) is larger than the pooled delta (+0.008, from 0.9385 to 0.9465), and the mean of the per-fold F1 values (0.9524) is above the pooled F1 (0.9465).
Verdict	Kept; became Stage 2 of the final pipeline.

A.9 Full pipeline, bias correction, and target reachability

Experiment A12: Full 5-fold training and pooled evaluation

Hypothesis	The winning fold-0 recipe (dihedral-aug detector + synthetic-aug detector, 8-view TTA + WBF decode, cross-fold verifier) will generalize across all five folds, and the pooled global F1 (the competition’s actual scoring method) will land close to the fold-0 number.
Method	Train the winning recipe on all five folds independently, decode all five with the frozen decode pipeline, apply per-fold verifiers, pool all 1,407 boxes across 200 pairs with one global threshold swept only on out-of-fold predictions.
Result	Pooled F1 0.9465, Precision 0.963, Recall 0.930 (TP 1,309 / FP 50 / FN 98). Per-fold F1: 0.943, 0.955, 0.935, 0.961, 0.968; fold 2 is the lowest, and fold 0 (the one all early experiments were tuned on) is the second lowest and below the per-fold mean of 0.9524.
Inference	The pooled out-of-fold F1 (0.9465) is higher than the fold-0 starting point (0.9049) mainly because of the ensemble, TTA and verifier added after Experiment 2. This is the number the submission’s decode threshold (0.43) was derived from; the threshold transferred to the test set (6.68 predicted boxes per image against 7.04 in training).
Verdict	Kept as final architecture.

Experiment A13: Nested cross-validation for threshold-selection bias

Hypothesis	Sweeping the global decode threshold on the same out-of-fold predictions being reported may inflate the reported F1; a nested CV check (threshold chosen on four folds, applied to the fifth) will quantify this optimism.
Method	For each held-out fold, sweep the threshold on the other four folds only, then score the held-out fold with that threshold. Compare to the threshold tuned on the full pooled set.
Result	Detector only: tuned 0.9385 vs. nested-honest 0.9385 (0 optimism). With verifier: tuned 0.9465 vs. nested-honest 0.9435 (+0.0030 optimism).
Inference	Threshold-selection bias is small and quantified, not a major source of the headline number’s optimism.

Verdict Adopted 0.9435 as the threshold-bias-corrected out-of-fold number. We now report three estimates in this order: official test score (0.9577), nested out-of-fold (0.9435), raw out-of-fold (0.9465).

Experiment A14: Checkpoint-selection bias measurement

Hypothesis Saving the best-scoring epoch on a fold’s own validation set and reporting that checkpoint’s score on the same data is equivalent to taking the max of several noisy measurements, inflating the true generalization estimate.

Method Compare each model’s `best.pt` (highest validation epoch) score against its final-epoch (`last.pt`) score, across the `_aug` and `_augsyn` training recipes.

Result `_aug`: mean gap +0.005 (range 0.000–0.011) — curves flat at the end, so selection costs little. `_augsyn`: mean gap +0.034 (range 0.020–0.058) — these models trained only 15 epochs and were still oscillating, so `best.pt` cherry-picks a lucky point.

Inference The `_augsyn` models carry meaningful checkpoint-selection bias because they were undertrained, not because the method is flawed; a longer schedule (e.g. 40 epochs) should close most of this gap. The official score (0.9577) is above both the raw out-of-fold F1 and the range we had derived from this analysis (0.925–0.94), so this analysis did not predict the test outcome; it measures a property of the out-of-fold estimate and is not a forecast of the test score.

Verdict Reported as a measured bias of the out-of-fold estimate. The out-of-fold pass was not re-run with `last.pt`.

Experiment A15: Recall ceiling and reachability analysis of the 0.97 target

Hypothesis Given the fixed candidate set, there is a hard upper bound on achievable F1 regardless of rescoring quality; this bound should be checked against the competition’s 0.97 target before investing further in the verifier/scoring branch.

Method Compute TP/FP/FN assuming a perfect rescorer (keep every true candidate, reject every false one) on the final candidate set.

Result The candidate set contains 1,352 of 1,407 boxes at IoU 0.5 → ceiling F1 = 0.9801 (TP 1,352, FP 0, FN 55), even for a flawless rescorer.

Inference 0.97 is below the ceiling (0.9801), so the target is not mathematically excluded. But even if every one of the 1,352 proposals that match a true box were kept, reaching 0.97 would need at most about 28 false positives (the out-of-fold pipeline has 50), and the achieved 0.9465 is 0.0336 below the ceiling. Most of the remaining gap to 0.97 would have to come from better scoring of a candidate set that already contains the 1,352 true boxes, with almost no false positives left, which is practically out of reach. The 55 boxes no model proposes at all are the real frontier; the sub-12 px band that used to hold most misses is now at 0.944 coverage, so that lever is largely spent. Further gains require better candidate proposals (a second

backbone, a properly trained `_augsyn`, direct inspection of the 55 missed boxes), not better verifiers.

Verdict Target judged practically out of reach with these proposals; a roadmap for the next phase was defined but not executed.

A.10 Silent implementation bugs

Three bugs were found during development. The first two corrupted fused scores and made working ideas (TTA, ensembling) look like failures until they were fixed; the third slowed verifier training and made a retrain expensive to attempt.

Experiment A16: WBF score saturation

Discovery An 8-view TTA run scored F1 0.775 at precision 0.67, worse than no TTA.

Cause Fused confidence was computed as $\text{sum}(\text{scores}) / \min(\text{members} + 1, 3)$. With 8 views a real box collects about 16 agreeing members, so the sum divided by 3 clipped to 1.0, while a box proposed by a single view kept its raw score; confidence was effectively inverted. The single-view path never saturated the denominator, so the bug appeared only once TTA or ensembling raised member counts. It invalidates every TTA comparison made before the fix. The numbers in Experiments 6 and later were produced after the fix; Experiments 3 and 4 predate it and were not repeated.

Fix Fixed by an agreement-scaled mean; the single-view path was verified bit-for-bit identical to the archived pre-fix number.

Experiment A17: Ensemble double-counted its sources

Discovery Pooling ensemble members collapsed recall to 0.056, which was easy to notice.

Cause Individual models already fuse their own TTA views internally, so defining $n_{\text{sources}} = \text{models} \times \text{views}$ over-divided every score by roughly 16×.

Fix Fixed by correcting the source-count accounting in the fusion step.

Experiment A18: Verifier data-loading inefficiency

Discovery The first verifier epochs took about 60 minutes with the GPU idle.

Cause The loader re-decoded four full-page PNGs per sample on every cache miss and shuffled across 160 pages, which meant about 140,000 PNG decodes per epoch. This affected throughput, not F1; it is not the cause of the null result in Experiment 10 (starved negatives), but it made the retrain of Experiment 11 expensive to attempt.

Fix Fixed by cutting crops once into memory upfront, which brought epoch time down to seconds.

A.11 Infrastructure Failure during Fold Training

A five-way parallel fold-training job on Modal died mid-run after a local network interruption. The fan-out (`starmap`) was driven by the client, so `SIGTERM` reached the whole

process group and killed all five folds even though `--detach` was set. Detach flags do not protect a client-orchestrated fan-out; we moved the fan-out into a remote function so that it survives local disconnects. This incident affected throughput only and no reported number.

B Task 2: Dataset, Full Experimental Log and Limitations

B.1 Dataset and Data Irregularities

Table 6 summarises the dataset. An exploratory pass (Experiment 1) found: folder RE-WA-2021-0077 actually contains *two* complete establishments (Goldfields #35 and Midwest #100) both internally stamped with the same RE number, explaining the “missing” 100th case; folders RE-QLD-2022-0077 and RE-SA-2021-0066 are missing Track 1 (registration form); 267 explicit [DEFICIENCY -- NOT DOCUMENTED] markers, concentrated in Tracks 2 and 6; subtler planted failures (records kept in Mandarin, violating English-records CPs; a “migration in progress” status where “2-year retention verified” was expected; worn door-seal gaps; single-operator dispatch checks); a Registered Commodity field randomized per document (~5.6 distinct values/case, generation noise rather than signal); and uppercase headings corrupted by a state-code find/replace bug (e.g. PHYTOSANITARY → PHYTOQLDNITARY, a generation artifact rather than a traceability failure).

Table 6. Task 2 dataset statistics.

Quantity	Value
Raw folders / logical cases (after manifest fix)	99 / 100
Files per case	9 tracks (7 .docx + 2 .xlsx)
Checking points	41 total: Element-1 (7), Element-2 (9), Element-3 (12), Element-4 (13)
Policy document	132 pages, 184 sections, ~266K characters
Labels	None provided; 10-case self-labelled development set; labels not independently verified
Median case size	~8.2K words / 22–26K tokens (fits in one context window)

B.2 Organisation of the Log

Fourteen numbered experiments were run; we group them by theme below. Within this appendix, “Experiment *n*” refers to Experiment *Bn*.

B.3 Exploratory data analysis and evidence engineering

Experiment B1: Exploratory data analysis of the 99 raw case folders

Hypothesis	The provided 99 folders correspond cleanly to 99 establishments with 9 consistent files each; before building any pipeline, this assumption should be checked for structural irregularities that could silently corrupt labels or ingestion.
Method	Enumerate every folder, count files per folder, diff file-naming patterns, spot-check document contents for internal consistency (e.g. RE number stamped inside each document versus the folder name).
Result	Found that folder RE-WA-2021-0077 actually contains two separate, fully documented establishments (Goldfields #35 and Midwest #100), both stamped internally with the same RE number, explaining why 99 folders needed to become 100 logical cases. Also found two folders missing Track 1, 267 explicit deficiency markers, and template-noise fields (randomized commodity field, a state-code find/replace bug corrupting headings).
Inference	A naive per-folder pipeline would have silently produced only 99 rows, misattributed the doubled establishment’s evidence to one RE number, and flagged generation artifacts as real traceability failures. The manifest layer and doubled-folder split are load-bearing, not optional cleanup.

Experiment B2: Canonical logical-case manifest and evidence ETL design

Hypothesis	A deterministic manifest layer that sits between the raw folders and the pipeline (rather than the pipeline touching raw folders directly) will make ingestion irregularity-proof and let the final row-shape (99 vs. 100 rows) be decided independently of ingestion logic.
Method	Define a canonical case model (case_id, folder_path, establishment name as identity anchor, farm_number, 9-track file map, irregularity flags). Ingest by sorted directory listing, mapping files to tracks by leading numeral, never asserting exactly 9 files present. Produce two candidate submission variants: Variant A (99 rows, doubled folder counted once) and Variant B (100 rows, doubled folder split with a provisional second ID).
Result	A case manifest and a logical-case list were produced; the ETL converts .docx/.xlsx per case into track-tagged markdown, preserving table structure (xlsx register rows carry decisive signals) and tolerating missing tracks.
Inference	Keeping compliance logic entirely out of the ETL layer (pure text/table extraction, no thresholds or rules) is what satisfies the competition’s “no hard-coded compliance logic” constraint at the data layer, in addition to the prompt layer.
Verdict	Kept; became Phase 2 (Deterministic Evidence ETL) of the final architecture.

B.4 Grounding: the Open Knowledge Format

Experiment B3: Open Knowledge Format (OKF) construction — CP-to-section grounding

Hypothesis	To satisfy the “no hard-coded compliance logic” rule while still giving the model focused legislative context, each of the 41 checking points should
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	be mapped to the specific policy sections that actually state the obligation (not just title-matching sections), and this mapping should be produced by an LLM reading the real legislative text rather than authored by hand.
Method	Chunk the 132-page policy PDF into 184 section files via a deterministic regex on section headers (PyMuPDF). Run a one-time LLM pass that reads the policy text and maps each of the 41 CPs to its governing section(s), producing a YAML-cited bundle. Deduplicate repeated citations.
Result	64 citation instances deduplicated to 18 unique governing sections (3.56× reduction). The bundle is stored with the system and can be re-derived on demand.
Inference	Because the mapping is machine-derived from the actual statute text rather than hand-authored rules, it satisfies the no-hard-coding constraint while keeping prompts short (18 sections instead of the full 184-section, ~266K-character document). The mapping is, however, model-dependent: only 12 of 41 CPs matched across two different mapping models, so the reproducibility contract is “same model + prompt → same mapping,” not a universal ground truth.
Verdict	Kept; became Phase 1 (OKF) of the final architecture.

B.5 Model and architecture selection

Experiment B4: Model bake-off across four frontier models

Hypothesis	Given identical prompts, OKF grounding, and full evidence, model choice (not architecture) will be the dominant factor in accuracy, and a reproducible, own-API-key-accessible model should be identified that comes closest to the best available model.
Method	Run identical full-context prompts + OKF + evidence for 10 dev cases across four models: Nemotron Ultra 550B, DeepSeek V4 Pro, Gemini 3.5 Flash, and Sonnet 5 (Claude).
Result	Nemotron Ultra: 314/410 = 76.6% (441K tokens, ~170s/case). DeepSeek V4 Pro: 334/410 = 81.5% (300K tokens, ~40s/case). Gemini 3.5 Flash: 353/410 = 86.1% (442K tokens, ~90s/case), with a +16pp Element-4 edge over Nemotron. Sonnet 5: 373/410 = 91.0% (~710K tokens, ~240s/case) — highest accuracy but withdrawn from use.
Inference	Gemini 3.5 Flash was selected as the best model that was both reproducible via the team’s own API key and cost/speed-competitive, even though Sonnet 5 scored 4.9pp higher; Sonnet’s exclusion is a reproducibility/access constraint, not a quality judgement. Sonnet 5 was used in this development bake-off and in the CP-level analysis of Experiments 9 and 10; the submitted system uses only Gemini 3.5 Flash.
Verdict	Kept — Gemini 3.5 Flash selected as the production model.

Experiment B5: Model tier vs. ensemble size (persona ensemble ablation)

Hypothesis	A 3-persona ensemble (Standard / Adversarial / Step-by-step, majority-vote arbitration) using a smaller/cheaper base model can match a single
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	strong-model pass, since ensembling is a standard way to compensate for a weaker base model.
Method	Run the 3-persona majority-vote ensemble with Haiku as the base model on a case; compare against a single Sonnet pass on the same case.
Result	Haiku ensemble: 70.7% on that case. Single Sonnet pass: 92.7% on the same case, at less token cost (70K vs. 163K).
Inference	On this case, the weak-model ensemble lost to a single strong-model pass (70.7% against 92.7%) while using more tokens. Model tier and ensembling are confounded here (Haiku ensemble against single Sonnet), so this suggests, but does not show, that majority voting cannot recover evidence that the individual personas never read.
Verdict	Dropped the persona ensemble in favor of a single strong full-context pass; whether ensembling would help with a competent base model was noted as an open question but not re-tested.

B.6 Retrieval versus full context

Experiment B6: Full-context vs. Hybrid RAG

Hypothesis	Retrieval-augmented generation [11], feeding the model only the most relevant evidence chunks per CP via BM25 [10] + dense retrieval fusion + ColBERT [12] late-interaction rerank, should reduce token cost relative to full-context while preserving or improving accuracy.
Method	Build a complete hybrid-RAG branch (BM25 lexical + bge-small dense retrieval [13], weighted fusion, ColBERT rerank, then LLM) and compare its 10-case accuracy and token cost against the full-context Gemini pipeline.
Result	Full-context (Gemini): 86.1% accuracy, 442K tokens. Hybrid RAG (best configuration, dense weight $\alpha = 1.0$): 74.9% accuracy, 610K tokens — worse on both axes.
Inference	When the full evidence already fits in one context window (median case ~22–26K tokens), retrieval only adds missed-signal risk and per-call overhead rather than saving cost. A concrete failure mode was measured: the decisive “maintained in Mandarin” sentence was retrievable only ~56% of the time under the RAG pipeline (a sentence-level rate; this is a different quantity from the track-level hit@5 of Experiment 7, which only asks whether the right evidence file is among the top five).
Verdict	Dropped hybrid RAG entirely in favor of full-context; reported as a high-value negative result since it overturned the initial retrieval-first design.

Experiment B7: Hybrid fusion-weight grid search (retrieval-quality-only, no LLM cost)

Hypothesis	Since RAG was already shown to underperform full-context, but the team still wanted to characterize retrieval quality in isolation: a moderate-to-BM25-heavy fusion weight was expected to perform best, matching the common prior that lexical retrieval anchors legal/structured text well.
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Method	Sweep the dense-retrieval fusion weight $\alpha \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$ against two embedding models (bge-small, bge-base) [13], measuring track-hit@5 against gold, with no LLM call involved.
Result	bge-small track-hit@5 by α : 0.515 ($\alpha = 0$, BM25-only) $\rightarrow$ 0.558 $\rightarrow$ 0.645 $\rightarrow$ 0.759 $\rightarrow$ 0.808 ($\alpha = 1.0$, dense-only). bge-base track-hit@5 at $\alpha = 1.0$ was 0.726, worse than bge-small's 0.808 at the same setting.
Inference	Pure dense retrieval ($\alpha = 1.0$) was optimal, inverting the prior that a higher BM25 weight would be better. At $\alpha = 1.0$ the larger embedding model (bge-base, 0.726) underperformed the smaller one (bge-small, 0.808); Table 9 gives the sweep.
Verdict	Recorded as a negative/inverted-prior finding; not used in production since full-context replaced RAG entirely.

B.7 Prompt and call-architecture engineering

Proposed separation of system and user prompts

Status. The submitted production run did not use a separate system prompt: the instructions were included in the user message together with the case context. The following system/user separation is documented as a proposed prompt-architecture revision, not as a configuration used to generate `submission_100.csv`. Any accuracy or robustness benefit remains unmeasured unless the revised configuration is evaluated experimentally.

Rationale. The proposed change separates stable auditor behaviour and output constraints from case-specific input. The system prompt contains the auditor role, verdict semantics, evidence-grounding rules, noise-resistance guidance, confidence requirements, and JSON output contract. The user prompt supplies the rendered context and checking points. This separation is intended to improve maintainability without adding CP-specific compliance criteria or hard-coded verdict logic.

Proposed system prompt. The proposed contents of `prompts/system_auditor.md` are reproduced below.

System Role: Evidence-Grounded Compliance Auditor.

You are a compliance auditor evaluating farm export-compliance cases against the *Export Control (Plants and Plant Products) Rules 2021*. Assess each checking point (CP) strictly against the governing policy text and case evidence supplied in the user message. Do not introduce requirements, thresholds, exceptions, or compliance criteria unsupported by the supplied text.

Verdict semantics. Assign exactly one verdict to each CP: 1 means the evidence establishes that the obligation is met; 0 means the evidence establishes that it is not met; and N/A is permitted only when the evidence gives an explicit, documented reason why the obligation does not apply. Silence or lack of discussion alone does not justify N/A.

Evidence-grounded reasoning. Identify the precise obligation in the supplied policy text, examine all relevant case evidence including narrative text, tables, registers, status

notes, and verification fields, and connect the decisive fact to the obligation. Do not invent evidence or infer a violation from suspicion alone. Do not overlook a deficiency merely because it appears in a dense table or an inconspicuous sentence.

Evidence integrity and noise resistance. Treat the establishment name and RE number in the case header as the authoritative identity anchors. Per-track registered-commodity fields and identity fields inside individual tracks may be independently generated template fields; do not treat mismatches as compliance failures unless the governing policy makes them relevant. Distinguish repeated boilerplate from case-specific evidence. Where a register contains a fixed target and a case-specific status or verification note, assess the complete row in context rather than treating the target as proof that the requirement has been met.

Grounding and scope. Use only the supplied policy text and case evidence. Do not invent legal citations, import outside requirements, or assume that a topically related section governs a CP. If the evidence does not resolve a factual question, state the limitation rather than manufacturing certainty. Do not introduce CP-specific verdict rules, fixed thresholds, or assumptions about expected labels.

Reasoning and confidence. For each CP, provide two to three concise sentences naming the governing requirement and the decisive case fact. Assign a confidence score from 0.0 to 1.0 that reflects the strength and clarity of the evidence. Include a short, verbatim excerpt from the case evidence; do not fabricate quotations or present paraphrases as direct quotations.

Output contract. Return one JSON object per CP in a single JSON array. Each object must contain exactly `cp`, `reasoning`, `verdict`, `confidence`, and `cited_evidence`. Include every requested CP exactly once. Return only valid JSON, without Markdown fences, introductory prose, trailing explanations, or additional fields.

Proposed user prompt. The dynamic user message is reduced to the following template; the existing placeholder construction remains unchanged.

```
# Compliance Audit Task
```

```
## Context
```

```
{CONTEXT_BLOCK}
```

```
## Task
```

```
Assess {CP_SCOPE_TEXT} against the supplied governing policy text and case evidence.
For each checking point, follow the system instructions and return the required
structured JSON verdict.
```

```
## Output
```

```
Return a JSON array containing one entry per checking point in scope.
```

Implementation note. This separation should be treated as a new experimental condition. To attribute any performance change to the system prompt, compare it against the existing single-user-message baseline using the same model snapshot, temperature, evidence, OKF bundle, and evaluation cases. Report results separately and do not describe the proposed prompt as part of the submitted run unless the production pipeline is rerun with this configuration.

Experiment B8: Token/cost engineering across prompt architecture versions (v1 → v2 → v3)

Hypothesis	Prompt/call architecture (how many LLM calls per case, and how evidence is chunked into those calls) can be iteratively simplified without losing accuracy, and simpler is likely both cheaper and more robust.
Method	v1: per-element, per-persona calls (13 calls/case, ~1.16M characters of combined prompt). v2: whole-case, 3-persona ensemble (3 calls/case, ~215K tokens). v3: single full-context call (1 call/case, ~44K tokens).
Result	v1 was unworkable due to megabyte-scale prompts. v2 surfaced a bug in which the majority-vote threshold was hard-coded at ≥ 2 , creating a hard-coded-logic risk and adding complexity. We did not measure a strong-model ensemble against a single strong pass (Experiment 5 compares different model tiers), so its accuracy benefit is untested. v3 (single full-context) became final: 1 call/case, ~44K tokens.
Inference	Architecture simplification tracked directly with the ensemble-vs-single-model finding (Experiment 5): once a capable model reads the full case in one pass, multi-call/multi-persona architectures add cost and failure surface; whether they add accuracy with a strong base model was not tested.
Verdict	Kept v3 (single full-context call) as final architecture.

Experiment B9: Prompt engineering fix for CP38 (record retention) — boilerplate-vs-signal discipline

Hypothesis	CP38 was being scored incorrectly because the model was anchoring on a boilerplate date-range string that is byte-identical across compliant and non-compliant cases, rather than on the actual per-case status note that indicates the real verdict.
Method	Diagnose failure cases: the model first always called CP38 compliant (reading only the date text), then, after a “do date arithmetic” prompt fix, always called it non-compliant (same accuracy, different wrong cases). Rewrote the prompt to explicitly instruct the model that the status/verification note (“2-year retention verified” vs. “migration in progress”) is the real signal, not the surrounding boilerplate date text. The exact added wording was: “Where a row has both a fixed target (e.g. ‘target $\geq$ 2 years’) and a case-specific status/verification note (e.g. ‘verified’ vs ‘migration in progress’ / ‘pending’), the status note is the signal — the target text next to it is the same in every case and is not.”
Result	CP38 accuracy on the 10-case dev set went from 4/10 to 10/10, which is +6 cells or +1.5 pp of the 410 judgments. Aggregate dev accuracy rose

	from 89.0% (365/410) to 91.0% (373/410), i.e. +8 cells, so two further cells, which we did not identify, also changed with the prompt rewrite. All of this was measured on the Sonnet dev run, whereas the submitted model is Gemini 3.5 Flash; we did not record whether the Gemini figure of 86.1% (Experiment 4) was measured before or after the CP38 fix. The CP38 fix was developed on the dev set it is scored on.
Inference	This generalized into a documented evidence-handling discipline in the production persona prompt: distinguish fixed boilerplate that repeats across compliant and non-compliant cases from the actual per-case status signal. We distinguish encoding a CP’s <i>requirement</i> (what the governing text obliges) from hard-coding its <i>expected answer</i> (instructing which verdict to emit for a specific evidence string). The added sentence is a general interpretation rule in structure, but its examples name the exact status strings that separate compliant from non-compliant rows, i.e. they are answer-specific to CP38; the no-hard-coding constraint is therefore only partially satisfied. The fix was evaluated only on the dev set it was tuned on, never on unseen cases. The same class of fix was identified as still needed, but not completed, for CP37.
Verdict	Kept; folded into the persona prompt’s three core evidence-handling disciplines.

B.8 Validation and error analysis

Experiment B10: Per-CP error analysis on the Sonnet dev run (diagnostic pass)

Hypothesis	Reviewing per-CP accuracy on the dev set (rather than only the aggregate) will surface both model failure modes and, potentially, defects in the hand-built gold labels themselves.
Method	Sort all 41 CPs by dev-set accuracy and manually review the worst-performing CPs’ evidence and gold labels.
Result	CP37 (50%): the model inconsistently flags a byte-identical boilerplate sentence as a deficiency where gold is unanimously compliant across all 10 cases — unfixed. CP39 (50%): gold itself is internally inconsistent — 4 cases share byte-identical evidence text but were labeled 1, 0, 0, 0 by the hand-labeler; the model’s own calls were judged more internally consistent than gold on this CP. CP35, CP36, CP2, CP40, CP1, CP6, CP7 and several 9/10 CPs were flagged as not yet root-caused. 32 of 41 CPs were at or above 90% accuracy, and 25 at 100% (16 CPs have at least one error).
Inference	Not every apparent model error is a model error — CP39’s low score reflects a gold-label defect rather than a reasoning failure, which matters for interpreting the ceiling on achievable accuracy given the dev set’s own label noise (the dev labels were not independently verified).
Verdict	CP37 fix identified as future work but not completed.

Experiment B11: Per-case variability on the 10-case dev set

Hypothesis	Dev accuracy should be reasonably stable across individual cases, rather than driven by one or two easy or hard cases.
Method	Compute accuracy for each of the 10 dev cases from the fixed Gemini 3.5 Flash predictions. Nothing is trained or held out in this analysis.
Result	Mean 86.1%, standard deviation 4.3 points across cases (not a confidence interval; for $n = 10$ the 95% interval of the mean is about ± 3.1 points); minimum case 80.5%, maximum 95.1%.
Inference	The official accuracy (78.18%) is below the worst dev case, so the official gap is not sampling noise within this dev set. Dev accuracy is not held-out accuracy: the CP38 part of the prompt was tuned on these cases (Experiment 9), the labels were self-assigned and not independently verified, and CP39's gold is internally inconsistent (Experiment 10). We therefore do not use 86.1% as a headline number.
Verdict	Retained as a description of per-case spread on the dev set, not as a validation estimate.

Experiment B12: Per-class metric analysis and the macro-vs-micro accuracy discovery

Hypothesis	Because the competition's scoring definition might use macro-averaged (per-class mean) accuracy rather than micro (overall) accuracy, per-class precision/recall/F should be computed explicitly to check whether the two scoring conventions would diverge materially.
Method	Compute per-class precision, recall, F1, and per-class accuracy for the three verdict classes (1/compliant, 0/non-compliant, N/A) on the 10-case dev set using Gemini 3.5 Flash's predictions.
Result	Class 1 ($n = 330$): precision 0.939, recall 0.885, F 0.911. Class 0 ($n = 72$): precision 0.616, recall 0.847, F 0.714. Class N/A ($n = 8$): precision 0.000, recall 0.000, F 0.000 — the model never emitted N/A at all, and all 8 gold-N/A cells (from the two Track-1-missing cases, on CP1/2/6/7) were predicted as 1. Micro-accuracy 86.1%, macro-F 0.54, macro-recall 0.58 $((0.885 + 0.847 + 0)/3)$.
Inference	On the dev set the model never predicts N/A, so a macro-averaged metric would give 0.58 against 0.86 micro. The official scores later showed that recall equals accuracy (Section 3.4), i.e. support-weighted averaging, so this concern did not materialise; we keep the analysis because it documents the N/A behaviour. The all-1 prediction on the two Track-1-missing cases is also an avoidable limitation: the ETL knows which track is missing, so a prompt rule “if the track is absent, answer N/A” would not be compliance logic (it concerns input availability). We did not apply it; at most 8 of 4,100 production cells (two cases $\times$ four CPs) are affected.
Verdict	Reduced to a limitation after the official results; the N/A behaviour is avoidable and was not fixed.

Experiment B13: Per-element accuracy breakdown (Gemini 3.5 Flash, dev set)

Hypothesis	Accuracy likely varies by compliance element (registration, facilities, hygiene, traceability), since these differ in how much they depend on dense tabular evidence versus narrative text.
Method	Compute accuracy per element (Element-1 through Element-4) on the 10-case dev set.
Result	Element-1 (registration/plans): 85.7%. Element-2 (buildings/facilities): 98.9%. Element-3 (hygiene/waste/pest): 85.8%. Element-4 (traceability/phytosanitary): 77.7%, the weakest element.
Inference	Element-4 (traceability), which depends most heavily on cross-referencing xlsx register rows and identifying subtle status-note signals (similar to the CP38/CP39 pattern), is the hardest element and the most likely source of remaining error in a production run.
Verdict	Documented as a known weak point, not separately re-engineered before the production run.

B.9 Production run

Experiment B14: Production run across all 100 logical cases

Hypothesis	The finalized single-call, full-context, Gemini 3.5 Flash pipeline, hardened with per-case checkpointing and a per-call timeout, will complete the full 100-case corpus reliably despite API rate limits and credit constraints encountered during earlier long runs.
Method	Run the final pipeline (Phase 1 OKF + Phase 2 ETL output + Phase 3 single full-context LLM call + Phase 4 formatting) across all 100 logical cases, with a 240-second per-call timeout and per-case checkpointing enabling resumption without lost progress.
Result	100/100 cases completed, 0 blanks, 0 parse failures after hardening. Total cost 4,493,869 tokens (~44.9K/case). Output distribution: $3,118 \times '1'$, $982 \times '0'$ (23.95%, against 17.6% in the dev gold), $0 \times 'N/A'$. Class-0 precision on the dev set was only 0.616, so over-flagging is a possible contributor to the gap between dev and official accuracy; without test labels we cannot confirm it. The run was resumed 4 times across mid-run key changes and credit/rate-limit stops, with zero lost progress due to per-case checkpointing.
Inference	The 100th case was submitted under a provisional ID. The N/A-avoidance behavior observed on the 10-case dev set (Experiment 12) persisted at full scale — zero N/A outputs across all 4,100 CP judgments in production, confirming this is a systematic model behavior rather than a small-sample artifact.
Verdict	Kept as final production run; the submission file was produced from this run.

B.10 Scorer Denominator

The official accuracy needs a denominator check: 0.7818×4100 is not an integer (3205 correct cells give 0.7817, 3206 give 0.7820), and 99 rows (4059 cells) does not give it either, so the scorer must exclude some cells or rows.

B.11 Additional Tables

Table 7. Task 2 model bake-off (10-case dev set, identical prompt and OKF grounding). Sonnet 5 was used for development only.

Model	Accuracy	Tokens	Time/case
Nemotron Ultra 550B	76.6% (314/410)	441K	~170s
DeepSeek V4 Pro	81.5% (334/410)	300K	~40s
Gemini 3.5 Flash (selected)	86.1% (353/410)	442K	~90s
Sonnet 5 (withdrawn: reproducibility)	91.0% (373/410)	~710K	~240s
Hybrid RAG, best configuration (Exp. B6)	74.9%	610K	—

Table 7 gives the model bake-off of Experiments 4 and 6, Table 8 the per-class breakdown of Experiment 12 and Table 9 the retrieval sweep of Experiment 7.

Table 8. Task 2 per-class metrics on the 10-case dev set (Gemini 3.5 Flash).

Class	n	Precision	Recall	F1
1 (compliant)	330	0.939	0.885	0.911
0 (non-compliant)	72	0.616	0.847	0.714
N/A	8	0.000	0.000	0.000
Micro-accuracy: 0.861		Macro-F1: 0.54		Macro-recall: 0.58

Table 9. Hybrid fusion-weight sweep (Experiment 7): track-hit@5 against gold for dense-retrieval weight α ($\alpha = 0$ is BM25 only, $\alpha = 1$ dense only). Only the $\alpha = 1.0$ point was reported for bge-base.

Embedding model	$\alpha = 0.0$	0.25	0.5	0.75	1.0
bge-small	0.515	0.558	0.645	0.759	0.808
bge-base	—	—	—	—	0.726

B.12 Limitations

- **Dev set.** Ten self-labelled cases with unverified labels; prompts, including the CP38 part, were tuned on them, so dev accuracy is not held-out accuracy.
- **Model mismatch in the analysis.** The CP37/38/39 analysis was run on the Sonnet 5 run, while the submitted model is Gemini 3.5 Flash.
- **No OKF ablation.** We did not compare with and without OKF, or OKF against the full 184-section policy, and only 12 of 41 CPs were mapped identically by two mapping models.
- **Ensembling untested for strong models.** Experiment 5 confounds model tier and ensembling.
- **Determinism.** Temperature 0 does not guarantee identical outputs, and the exact model snapshot is not recorded.
- **CP-specific prompt text.** The CP38 wording is a general interpretation rule whose examples name the exact CP38 status strings (“2-year retention verified” vs. “migration in progress”), so it is answer-specific and sits in tension with the no-hard-coding constraint; it was never evaluated on unseen cases, so the constraint is not fully demonstrated.

C Task 3: Dataset, Full Experimental Log and Limitations

C.1 Dataset

Table 10 summarises the data.

Table 10. Task 3 dataset statistics.

Quantity	Value
Training / test flows	1,285 / 327
Classes	10 (5 applications $\times$ voice/video)
Class sizes	40 (GoogleMeet_voice) to 256 (Discord_video)
Raw fields per flow	5 packet lengths, 5 relative timestamps
Packet lengths	26–1,242 bytes
Five-packet time span	48 μ s to 23.21 s (median 571 μ s)
Source calls (training)	$\approx$ 400 (estimated, see Exp. 2; identifiers not released)
Train/test adversarial-validation AUC	0.529 (exchangeable)
Constant-size flows	125 (Discord: 58 voice, 37 video)
Audio-only windows (all five packets < 300 B)	726 flows (56.5%)
Zoom audio-only windows	160 (80 voice, 80 video)
Official metrics	accuracy, precision, recall, F_β (weighted)
Development metric	macro-F1

C.2 Organisation of the Log

Twenty-five numbered experiments were run in a logical sequence; we group them by theme. Within this appendix, “Experiment n ” refers to Experiment Cn . Unless noted,

scores for Branch B are grouped cross-validation scores (Experiment 2). Branch A was developed earlier, under stratified 5-fold cross-validation with nested tuning, and its numbers are labelled accordingly (see Section C.11). Metrics are accuracy, macro-F1 and, for Branch A, macro-recall.

C.3 Data and validation protocol

Experiment C1: Exploratory analysis of the flow records

Hypothesis	Before any modelling, structural properties of the 5-packet records (duplicates, train/test shift, label determinism, protocol markers) decide how much is learnable and how evaluation must be built.
Method	Integrity checks; per-column Kolmogorov–Smirnov tests and adversarial validation (a RandomForest separating train rows from test rows); per-class packet-length and inter-arrival analysis against protocol reference bands; mutual information and ANOVA- F per engineered feature; embeddings coloured separately by application and by call mode; flat versus factorised (application $\times$ mode) classifiers.
Result	1,285 training and 327 test flows, no missing values, <code>relative_time_0</code> identically zero. Adversarial-validation AUC 0.529. Only 2 of 1,136 distinct length tuples map to more than one class, but they cover 95 rows (7.4%). Packets in the 1100–1600 B band make up 10.7% of video-class packets and 0.0% of voice-class packets. Factorised, the application is recoverable at 0.912 and the call mode only at 0.888. Flows per call extrapolated to 100 held-out calls predict ≈ 321 test flows against 327 supplied.
Inference	Train and test are exchangeable, so cross-validation is a meaningful proxy. Call mode, not application, is the harder factor. Because most length tuples are unique, tuple-to-label determinism says little about held-out rows until it is tested on them (Experiment 13).
Verdict	Kept as the standing diagnostic; motivated the feature bands and the validation design.

Experiment C2: Plain versus grouped validation, and negative controls

Hypothesis	The 1,285 training flows come from an estimated 400 source calls and no call identifier is released, so near-identical sibling flows will straddle the train/validation boundary under ordinary stratified folds and may inflate scores. If folds and cached features are leak-free, shuffled labels must score near chance and a prior-only predictor must match the majority share.
Method	Two schemes were run side by side for every Branch B model: <i>plain</i> (repeated stratified 5-fold, 3 repeats) and <i>grouped</i> (StratifiedGroupKFold, 5 splits, 2 shuffled repeats = 10 folds) over a pseudo-group key equal to the packet-length tuple rounded to 8-byte granularity joined with $\text{round}(\log_{10} \text{duration}, 1)$. Fold indices were stored. Controls: N1, grouped CV with permuted labels; N2, prior-only predictor.
Result	The key yields 1,107 pseudo-groups; 253 flows (20%) share a group with at least one other flow. Plain and grouped scores of matched runs differ by

	up to about two points, and not always in the same direction (e.g. Experiments 10 and 14). N1 scores 0.1549 (expected band 0.08–0.20); N2 scores 0.1992 (expected 0.1992).
Inference	Sibling leakage is moderate at this data size but cannot be ruled out per model, so grouped scores are the ones every selection decision relies on. Folds and feature cache are leak-free. Two caveats. First, the pseudo-key is conservative in one direction only: 1,107 pseudo-groups against an estimated 400 calls means the key splits some calls, so grouped CV may still leak siblings (Experiment 15 finds 1,024 singleton clusters). Second, for TabICL at full width the grouped score (0.8389) is <i>higher</i> than the plain score (0.8195), the opposite of what sibling leakage predicts; we have not isolated the cause and treat the two protocols as differing by seed and fold structure as much as by leakage.
Verdict	Grouped CV adopted as the decisive estimate; both controls pass.

C.4 Baselines, model search and features

Experiment C3: Untuned baseline and large-scale model search

Hypothesis	A broad search across model families finds a configuration that clearly outperforms an untuned gradient-boosted baseline.
Method	Untuned LightGBM on the engineered features (plain 5-fold) as control. Then 2,377 points scored (XGBoost 962, LightGBM 626, RandomForest 588, CatBoost 61, ExtraTrees 36, SVM 27, MLP 23, k -NN 22, HistGradientBoosting 20, logistic regression 10, PacketFormer 2) in 83.6 container-hours, one container per grid point, then Optuna [15] (TPE, ask/tell batching) around the grid winners. Class weighting was itself a searched parameter. Grouped CV.
Result	Untuned LightGBM 0.810 accuracy (plain protocol, so not directly comparable with the grouped scores below). Optuna: LightGBM 180 trials $\rightarrow$ 0.8249, XGBoost 180 $\rightarrow$ 0.8202, RandomForest 150 $\rightarrow$ 0.8023; the CatBoost [16] study stopped when a single trial exceeded the 3,600 s limit. Table 11 gives the best point per family. The winning LightGBM point uses 800 trees, 31 leaves, learning rate 0.12, subsample 0.8, column sample 0.8, minimum child samples 10 and no class weighting.
Inference	Boosted trees lead; distance, kernel, linear and neural models trail by 9–11 points. The untuned baseline was scored under the plain protocol and the searched points under grouped CV, so the gain attributable to the search is not isolated; the grouped-CV gap between the tuned LightGBM (0.8249) and the default-setting LightGBM of Experiments 4 and 17 (0.8152, 0.8156) is about one point.
Verdict	LightGBM (0.8249 grouped, macro-F1 0.8069) kept as the control for all later Branch B comparisons. The “LightGBM control” appears with four scores in this section (0.8249 here; 0.8152 in Experiment 4; 0.8210 in Experiment 14; 0.8156 in Experiment 17) because they come from different configurations and repeat counts; Table 12 lists them.

Table 11. Best point per family, grouped CV (Experiment 3).

Rank	Model	Grouped accuracy	Macro-F1
1	LightGBM	0.8249	0.8069
2	XGBoost	0.8218	0.8067
3	HistGradientBoosting	0.8156	0.7959
4	ExtraTrees	0.8125	0.8009
5	CatBoost	0.8070	0.7895
6	RandomForest	0.8023	0.7866
7	PacketFormer (pretrained)	0.7346	0.7262
8	PacketFormer (scratch)	0.7331	0.7239
9	MLP	0.7292	0.6792
10	SVM (RBF)	0.7268	0.6595
11	Logistic regression	0.7144	0.6666
12	k -NN	0.7132	0.6747
–	Majority class	0.1992	0.0332

Table 12. The four LightGBM scores used as “control” in this section.

Experiment	Configuration	Grouped accuracy
3	Optuna best point, 241 columns	0.8249
4	matched full-width run, 241 columns	0.8152
14	best point used for calibration	0.8210
17	standing configuration, three repeats	0.8156

Experiment C4: Feature width and label-aware blocks

Hypothesis	Pruning the 241 deterministic columns by mutual information should remove noise at 1,285 rows; class-conditional Markov, density and neighbourhood features computed inside the CV loop should add signal beyond the label-free blocks.
Method	Matched LightGBM and XGBoost runs on the full 241 columns, top-120 and top-60 columns, full repeat budget; then matched runs with the label-aware blocks switched on and off.
Result	LightGBM grouped accuracy 0.8152 (full), 0.8097 (top-120), 0.7895 (top-60); macro-F1 0.7974, 0.7918, 0.7706. XGBoost 0.8016 (full) against 0.7907 and 0.7911. Label-aware blocks: LightGBM 0.8152 $\rightarrow$ 0.7969, XGBoost 0.8016 $\rightarrow$ 0.7883, at every width tested.
Inference	The full set beats both subsets in every matched pair, so pruning discards usable signal. At 1,285 rows the label-aware blocks contribute variance rather than signal (about 1.8 accuracy points on LightGBM) even after the inner out-of-fold fix.
Verdict	Full 241-column label-free set kept for Branch B; label-aware blocks dropped.

Experiment C5: Compact 31-feature set and two added features (Branch A)

Hypothesis	Branch A needs only a compact representation (raw lengths and times, differences, order statistics, shape descriptors); single features found by
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	symbolic search or by inspecting misclassified Discord and Google Meet video flows may still add signal.
Method	29 base columns (Stage 0 of Section 4.2) plus two features added one at a time: log of the fourth packet length (found by genetic-programming symbolic regression), and the position of the largest packet-to-packet size increase (separating an instant jump to a plateau from a gradual ramp, found by manual inspection of video misclassifications).
Result	An isolated gain in the validated score of +0.0017 for the log feature and +0.0014 for the jump position. Total 31 features.
Inference	Small but individually reproducible gains; the jump position encodes the temporal shape that static order statistics miss.
Verdict	Both kept; Branch A uses 31 features.

C.5 Branch A: tree ensemble

Experiment C6: Probability correction on the averaged ensemble

Hypothesis	Averaged tree probabilities are over-confident and carry the training class prior; softening them with a temperature and dividing out the prior should help a macro-averaged criterion.
Method	Out-of-fold probabilities from the three models (Stage 1 settings) under stratified 5-fold (seed 42); equal-weight average; temperature T on the log-probabilities; division by the empirical class prior; argmax. T was fixed at 2.0 after the temperature studies of this branch.
Result	Stage-0 macro-recall 0.8286.
Inference	Reference point for the decision layer of Experiment 7.
Verdict	Kept.

Experiment C7: Per-class decision bias and margin-triggered specialist

Hypothesis	Four classes (Zoom_video, GoogleMeet_video, Zoom_voice, Discord_video) were identified as weak, and the residual confusions among five classes (GoogleMeet voice/video, Discord voice/video, Messenger_video) are genuine two-way ties with a sample-dependent partner. An additive log-probability offset on the weak classes, plus a specialist model for tied cases, should recover them. Zoom_video is the class sacrificed, so leaving its offset unconstrained should be harmful.
Method	Within five outer folds, offsets for the four weak classes were tuned by coordinate ascent on the tuning rows only (coarse grid -1.5 to 1.5 in 31 steps, then a ± 0.15 fine grid, up to 10 passes). A specialist LightGBM trained only on the five confusable classes replaced the prediction whenever the top-1 and top-2 classes were both confusable and the probability margin was below $\tau \in \{0, 0.05, \dots, 0.25\}$; τ and the specialist were fitted in an inner four-fold loop. Final constants are medians across outer folds. The Zoom_video offset was also capped at ≥ -0.10 and compared with unrestricted optimisation.

Result	Macro-recall 0.8286 $\rightarrow$ 0.8379 (nested). Final offsets: Zoom_video -0.10 , GoogleMeet_video $+0.20$, Zoom_voice $+0.12$, Discord_video $+0.38$; $\tau = 0.05$. Uncapped Zoom_video offset: 0.8358; capped: 0.8379.
Inference	A gain of 0.0093 over the corrected ensemble, obtained by shifting the decision boundary toward recall (the precision cost is measured in Experiment 8). Unconstrained optimisation pushes Zoom_video further down for a worse net result.
Verdict	Kept as the stand-alone Branch A configuration, with the cap.

Experiment C8: F1 view of Branch A

Hypothesis	Recall-oriented offsets improve macro-recall at some cost in precision; macro-F1, the scored metric, will show where.
Method	Macro-, micro- and weighted-F1 and per-class recall and F1 on the nested predictions of Experiment 7.
Result	Macro-F1 0.8033, micro-F1 0.8078, weighted-F1 0.8131, against macro-recall 0.8379. Zoom_voice recall 0.800 but F1 0.549; GoogleMeet_voice recall 0.950, F1 0.724; Zoom_video recall 0.552, F1 0.691; WhatsApp_voice recall 0.988, F1 0.975.
Inference	Zoom_voice precision is about 0.42: the offsets buy recall with false positives, so the macro-recall figure overstates the criterion actually scored.
Verdict	The branch is reported on both metrics; its outputs enter the blend as corrected probabilities <i>without</i> the offsets or the specialist (Experiment 25).

Experiment C9: Single-thread determinism

Hypothesis	Parallel fitting should not change which hyperparameter pair the inner search selects.
Method	Estimators were run with <code>n_jobs=-1</code> and <code>n_jobs=1</code> ; the selected (temperature, margin-threshold) pair and the final score were compared.
Result	Parallel execution introduced floating-point non-determinism that shifted the selected pair and moved the final score by up to about 0.01.
Inference	Differences of this size are as large as several of the gains reported in this section.
Verdict	All estimators run with <code>n_jobs=1</code> .

C.6 Branch B: tabular in-context learning

Experiment C10: TabICL against the tree control

Hypothesis	A pretrained in-context tabular classifier is competitive on 1,285 rows without a hyperparameter search.
Method	TabICL at default settings on the full 241 columns, top-120 and top-60, against the LightGBM control of Experiment 3; paired McNemar test [17]. TabPFN v2 [18] could not be run initially because its weights require licence acceptance through an interactive login; it was later run on the published checkpoint.

Result	Full width 0.8389 grouped (plain 0.8195, macro-F1 0.8232, 83 s); top-120 0.8280 (0.8117); top-60 0.8272 (0.8117); LightGBM control 0.8249 (0.8069). Against the control TabICL wins 66 rows and loses 49 ($p = 0.135$). TabPFN v2: 0.8311 (top-120), 0.8230 (full), 0.8191 (top-60), degrading with width.
Inference	All three TabICL widths exceed the best of 2,377 searched points with three arms and no tuning, but the paired difference is not established at this sample size. TabICL is also the best foundation model tried.
Verdict	Adopted as the second branch on the strength of the point estimate and its consistency across widths.

Experiment C11: TabICL parameters and seed bagging

Hypothesis	The internal ensemble size, the softmax temperature and bagging over seeds can add accuracy cheaply.
Method	Eighteen arms over <code>n_estimators</code> , <code>softmax_temperature</code> and bag count, full repeat budget.
Result	<code>n_estimators=16</code> : 0.8405 grouped (macro-F1 0.8268), +0.0016 over the default 8 against a noise floor of 0.0031; 32: 0.8358. The softmax temperature (0.75/0.90/1.05) gives identical accuracy to four decimals. Five seed bags of 8 estimators: 0.8350 (sd 0.0016) against 0.8389 unbagged (sd 0.0031).
Inference	The internal ensemble has an interior optimum; temperature cancels out of the argmax under logit averaging and acts as a calibration parameter only; bagging halves the variance but smooths toward a slightly worse consensus because the model already ensembles internally.
Verdict	None of the three changes is established on accuracy grounds; the runs that feed the blend nevertheless use 16 estimators.

C.7 Routes that did not help

Experiment C12: Auxiliary corpus: pretraining, probe features and direct transfer

Hypothesis	A small transformer over the five packets, pretrained on a larger public capture corpus that contains all five applications, transfers representations that the trees cannot learn from 1,285 rows; frozen-encoder probes add information absent from the engineered features; and a classifier trained purely on the corpus recognises the application on competition flows if the two corpora are commensurable.
Method	The corpus (MIRAGE-AppAct-2024 [21], 6.97 GB, 20 applications) was reduced to payload lengths, gaps and direction for the first 20 packets: 14,975 UDP flows, or 90,610 flows with TCP included (the larger set was adopted, with a transport flag). <i>Pretraining</i> : PacketFormer tokenises each packet (eight continuous descriptors, a 24-bin size embedding, positional embedding, [CLS]); 3 pre-norm layers, 4 heads, $d_{\text{model}} = 96$; 8,000 steps, batch 256, 347 s on one GPU; masked-packet modelling, a contrastive objective over two traffic-plausible augmentations, a domain-adversarial

head against unlabelled competition flows and random truncation to 5/8/12/20 packets. No competition label was used. *Probes*: 46 probe columns (application family, a video-likeness proxy, bitrate and pacing with residuals, a transport fingerprint, PCA components) appended to the full feature set; matched runs with the block on and off, for LightGBM, XGBoost, RandomForest and TabICL at 8, 16 and 32 estimators. *Transfer*: a gradient-boosted model trained on 7,655 UDP media flows of the five applications was applied to the competition training set, with lengths as stored and with 28 bytes of IP+UDP header added; the corpus metadata was inspected for an activity label.

Result	Pretraining: from scratch 0.7331 grouped accuracy (macro-F1 0.7239), pretrained 0.7346 (0.7262), a gain of 0.0015; final pretraining losses: total 2.664, masked-bin 1.146, masked-gap 0.959, contrastive 2.074, domain 0.016. Probes (grouped accuracy): LightGBM 0.8144 $\rightarrow$ 0.8144, XGBoost 0.8082 $\rightarrow$ 0.7996, RandomForest 0.7778 $\rightarrow$ 0.7782; TabICL with probes 0.8257 / 0.8288 / 0.8300 against 0.8389 (8) and 0.8405 (16) without, best arm against best arm -0.0105 ; the transport-fingerprint probe had a degenerate target and emitted no features. Transfer: application accuracy 0.3183 and 0.2988 respectively, against 0.3844 for the majority class; the published archive carries only the package name and byte/packet counters, so no activity manifest exists.
Inference	Five tokens give attention little to work with and 1,285 labels give fine-tuning little to learn from. The probe block is neutral-to-harmful for three model families and marginal for a fourth. Transfer falls below the majority baseline: the domain gap at the level of packet sizes is total, which also explains the null probe result.
Verdict	Auxiliary-data route closed. PacketFormer kept only as a diversity member of the model pool; probe block retired.

Experiment C13: Exact and tolerance-based length-tuple lookup

Hypothesis	Because the label is almost determined by the length tuple (Experiment 1), a lookup table over training tuples can override the model on rows it matches.
Method	Coverage and purity of tuple matches at exact, ± 1 , ± 2 and ± 4 byte tolerance (gate: more than 5% coverage at more than 0.95 accuracy), followed by the differential that decides the question: model accuracy against lookup accuracy on the covered rows.
Result	Exact matching covers 7.6% of test rows (5.5% of CV rows) at 0.972 accuracy and passes the gate. On the 71 covered CV rows the model is already at 1.0000 against the lookup's 0.9718; overall accuracy with the override changes by -0.0016 . At ± 4 bytes: 232 rows, model 0.9181, lookup 0.8922, change -0.0047 .
Inference	The model is already perfect on every exactly matched row; a lookup can only take rows from a model that answers them correctly. The near-determinism seen in Experiment 1 is a property of unique tuples, not of exploitable structure.

Verdict Dropped.

Experiment C14: Calibration and decision-layer adjustments

Hypothesis	Confidence can be repaired without moving accuracy; a per-class logit bias, or deferring low-margin rows, lifts accuracy, particularly inside the Zoom pair.
Method	Temperature [19], vector and Dirichlet [20] calibration on the grouped out-of-fold predictions of the best LightGBM point; the selection rule was amended mid-round to the best negative log-likelihood among calibrators within one point of raw accuracy, because the first rule selected Dirichlet and silently gave up accuracy. Nested logit-bias fits on raw and on calibrated probabilities, and margin-based deferral at 0.10 and 0.35.
Result	Raw: accuracy 0.8210, NLL 0.7931, ECE 0.1012, mean confidence when wrong 0.7833. Temperature: 0.8210, 0.5628, 0.0451, 0.6095. Vector: accuracy 0.8031, NLL 0.5898, ECE 0.0359. Dirichlet: 0.8101, 0.5450, 0.0349. Decision layer: raw argmax 0.8210 (Zoom-pair accuracy 0.6548); bias on raw 0.8226 (5 rows changed); bias on calibrated 0.8187 (18 rows); deferral 0.8202 at 0.10 and 0.8117 at 0.35. The fitted bias vector is small (max $ b = 0.10$) and does not push toward Zoom_video.
Inference	Temperature scaling cuts NLL by 29% and ECE by 55% at no accuracy cost; the richer calibrators buy slightly more calibration for 1–2 accuracy points. The best decision-layer arm moves five rows for +0.0016, inside the noise floor; a global per-class prior shift cannot recover the Zoom-pair gap, because buying Zoom rows costs rows elsewhere.
Verdict	Temperature scaling kept; decision-layer adjustments dropped.

Experiment C15: Recovering call structure from the flows

Hypothesis	If flows from one call can be grouped, aggregating predictions within a group should resolve ambiguous flows from their siblings.
Method	A same-call scorer, clustering of its output, and aggregation of per-flow predictions within clusters by log-probability sum or confidence-weighted vote, plus an oracle run using the pseudo-groups as if they were true calls.
Result	Same-call ROC-AUC 0.998 on held-out groups; cluster application purity 0.997, label purity 0.989, adjusted Rand index 0.775 against the pseudo-groups; 1,024 of 1,107 clusters are singletons, the largest holds 13 flows. Aggregation: 0.8327 (log-probability sum, -0.0054) and 0.8304 (weighted vote, -0.0078) against 0.8381; oracle grouping 0.8311 (-0.0070).
Inference	Call structure is easy to recover and worthless: even with perfect knowledge of which flows share a call, aggregation lowers accuracy, because siblings are rare and where they exist they already agree with the model.
Verdict	Branch closed on a measured ceiling.

Experiment C16: Combining models inside Branch B, and selecting the Branch B model

Hypothesis	Combining diverse learners beats the best single model; a weighted combination of TabICL and a tree model, or a larger pool, beats the pre-specified TabICL pair.
Method	Hill-climbed soft-vote weights (also scored nested: fitted on four fifths of the out-of-fold rows, scored on the fifth), stacking with a logistic meta-learner (inner CV) and rank averaging, in two rounds: the gradient-boosted pool, then the pool with TabICL. Then 17 candidate blends over TabICL and the five other families on the same grouped folds with and without the Zoom rule, with paired bootstrap and McNemar comparisons against the earlier standing model (TabICL, flat view only). The pre-specified pair (flat and hierarchical views averaged) was kept as the default, because the best-scoring blend is the maximum of seventeen candidates scored on the same 1,285 rows.
Result	Round one: best single LightGBM 0.8210; nested hill-climb 0.8195; stacking 0.8187; rank averaging 0.7735; the fitted hill-climb score 0.8319 is 1.2 points above its nested score. Round two: single TabICL 0.8381; stacking 0.8327; nested hill-climb (TabICL $\times 0.75$, LightGBM $\times 0.25$) 0.8311; fitted 0.8451; rank averaging 0.7837. Candidate blends, accuracy / macro-F1 without the rule, and macro-F1 with it: TabICL $\times 2$ + LightGBM (flat+hier) 0.8381 / 0.8216, 0.8319; TabICL flat+hier average 0.8358 / 0.8201, 0.8292 ; TabICL flat only 0.8381 / 0.8262, 0.8274; TabICL + LightGBM + XGBoost 0.8342 / 0.8147, 0.8231; all six families 0.8272 / 0.8082, 0.8172. Shipped against standing: +0.0030 macro-F1, interval $[-0.0130, +0.0201]$, $P(\Delta > 0) = 0.630$, McNemar 46/51 ($p = 0.685$); observed maximum against standing: +0.0058, $[-0.0114, +0.0231]$, 0.744, 53/54 ($p = 1.000$).
Inference	At 1,285 rows a fitted blend weight is a memorised weight: only nested estimates are admissible for choosing. Neither paired difference in model selection is established; the durable result of the round is the diagnosis of which flows are unwinnable (Experiment 18), not the choice among near-equal blends.
Verdict	TabICL flat and hierarchical views averaged, with the Zoom rule (grouped accuracy 0.8342, macro-F1 0.8292), is Branch B.

C.8 Error geometry and the Zoom rule

Experiment C17: Where the errors are

Hypothesis	The residual errors concentrate in a small number of identifiable confusions rather than being spread evenly.
Method	Three-repeat grouped CV of the standing gradient-boosted configuration; errors split by label dimension and by packet-size regime (whether all five packets lie in the audio band, below 300 bytes).
Result	Accuracy 0.8156, macro-F1 0.7982, 237 errors. Application accuracy 0.9035; mode accuracy 0.8934 (0.9027 given the application is right). Audio-only windows: 726 flows (56.5%) at 0.7590 with 175 errors; windows with a larger packet: 559 flows at 0.8891 with 62 errors. Zoom_voice against

Table 13. Audio-only windows by application (Experiment 18). The four applications shown account for 646 of the 726 audio-only flows.

Application	Audio-only flows of which video		Grouped AUC (video vs voice)
Discord	290	52	0.880
Google Meet	85	45	0.826
Messenger	111	19	0.879
Zoom	160	80	0.551

Zoom_video alone accounts for 68 of the 237 errors; Zoom_voice recall is 0.34.

Inference Errors concentrate in audio-only windows and in one pair.

Verdict Redirected the work from model search to the Zoom pair.

Experiment C18: Separability of voice and video inside the audio-only window, and the score ceiling

Hypothesis A video call whose first five packets are all audio-sized produces a record indistinguishable from a voice-call record; whether the pair is separable depends on the application.

Method Per application, over flows whose window is audio-only, grouped-CV AUC of video against voice (gradient boosting over all 241 features); threshold sweep of the audio band from 200 to 500 bytes; upper bounds on accuracy and macro-F1 assuming every other flow is classified perfectly.

Result Table 13. For Zoom the subset is exactly balanced (80 voice, 80 video) and the AUC of 0.551 is not distinguishable from a coin flip at $n = 160$. Every threshold from 250 to 500 bytes isolates the same 80/80 subset (75/76 at 200 bytes). Upper bounds: accuracy 0.9377 (80 of 1,285 flows unwinnable) and macro-F1 0.9364, reached by sending the whole subset to Zoom_voice (Zoom_voice F1 0.667, Zoom_video F1 0.697). A macro-F1 of 0.90 would need the other eight classes to average $F1 \geq 0.954$; they average 0.868.

Inference The Zoom pair is information-limited by the five-packet window, not by the model; the 300-byte boundary is a property of the data, not of a tuned constant. This is the Task 3 counterpart of Task 1's perfect-rescorer ceiling (Exp. A15): the headline metric cannot go past 0.9364, so the remaining distance is in the other classes.

Verdict A macro-F1 of 0.90 is not attainable on this dataset; the quantity of interest becomes how much of the structural maximum is captured.

Experiment C19: Derivation and measurement of the Zoom rule

Hypothesis On an evenly split, inseparable subset every assignment has the same accuracy but not the same macro-F1; sending the whole subset to the smaller class recovers its recall at no cost to the larger one.

Method Derivation: with all other flows correct, sending the 160 audio-only Zoom flows to Zoom_voice gives F1 0.667 (voice) and 0.697 (video; recall 92/172), summing to 1.364, whereas sending them to Zoom_video gives 0 and

Table 14. Macro-F1 per family without and with the Zoom rule (Experiment 20); each column reports the best of the three views, chosen separately per column, so the two columns are not a matched-view comparison.

Base learner	Without rule	With rule
TabICL	0.8262	0.8292
LightGBM	0.8066	0.8169
XGBoost	0.7965	0.8044
ExtraTrees	0.7815	0.7885
CatBoost	0.7735	0.7804
RandomForest	0.7640	0.7725

0.811, summing to 0.811. Rule: if a flow is predicted to be Zoom (either class) and all five packets are below 300 bytes, label it Zoom_voice. The rule has no fitted parameter: its direction follows from the 80/80 balance and the class sizes, and its threshold is the audio band already defined from the protocol analysis. Out-of-fold evaluation on the flat, hierarchical and averaged views of the round-three base learner; bootstrap over 2,000 re-samples of the averaged model.

Result	Macro-F1 0.7942 $\rightarrow$ 0.8080 (flat), 0.7978 $\rightarrow$ 0.8092 (hierarchical), 0.8041 $\rightarrow$ 0.8181 (average). Bootstrap gain on the average: +0.0140, 95% interval [0.000, +0.028], $P(\text{gain} > 0) = 0.975$; accuracy 0.8233 $\rightarrow$ 0.8241; Zoom_voice F1 0.384 $\rightarrow$ 0.580.
Inference	The gain comes from Zoom_voice recall, with accuracy essentially unchanged, as the derivation predicts. The bootstrap interval touches zero at its lower end. The rule was derived for macro-F1; on accuracy it is neutral (0.8233 $\rightarrow$ 0.8241), so it should not be expected to move an accuracy-type score.
Verdict	Kept as a deterministic post-processing step.

Experiment C20: Does the rule hold across model families?

Hypothesis	A rule derived from a class-balance measurement, not fitted, should help every hypothesis class.
Method	Six base learners, each in three views (flat ten-class, hierarchical, fifteen-class mixture in which video is split by window type), grouped folds, with and without the rule.
Result	Table 14: all six families improve, in every view. For TabICL the matched flat+hierarchical average goes from 0.8201 to 0.8292 (+0.0091); the +0.0030 implied by the table's best-view columns understates it.
Inference	The rule reflects the data rather than the estimator.
Verdict	Kept.

Experiment C21: Fitted corrections against the derived rule

Hypothesis	Hierarchical heads, a finer label view, balanced weights or a fitted decision layer add to the rule.
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Method	Each change compared on grouped out-of-fold predictions; fitted parameters scored nested.
Result	Hierarchical application $\times$ mode heads: +0.004 macro-F1 alone and a useful second view. Fifteen-class mixture view: 0.8063 alone, above the flat view. Average of flat, hierarchical and mixture: 0.8085, 0.8185 with the rule. A single global mode head in place of per-application heads: AUC 0.9661 against 0.9680, no difference. Class-balanced sample weights in every head: flat +0.005, ensemble -0.003 . Per-class log-probability bias fitted for macro-F1: in sample 0.8124, nested 0.7956 against 0.7982 raw. A single global mode-bias parameter: nested 0.8173 against 0.8181 unbiased.
Inference	Both fitted decision-layer variants gain in sample and lose nested; at 1,285 rows a fitted threshold is a memorised threshold, which is why the derived rule survives where fitted corrections do not.
Verdict	Hierarchical and mixture views kept as ensemble members; weights and fitted biases dropped.

C.9 Combining the two branches

Experiment C22: Paired out-of-fold diagnostics

Hypothesis	The two branches, built on different features (31 against 241 columns) and different learner families, make partly different errors, so blending can help.
Method	Raw Branch A probabilities (temperature $T = 2$, prior-corrected) against the averaged Branch B flat and hierarchical probabilities on the same 1,285 training rows.
Result	Macro-F1 / accuracy: Branch A 0.7900 / 0.7961, Branch B 0.8201 / 0.8358. The two agree on 84.98% of rows; both are correct on 75.33% and both wrong on 12.14%; only Branch A is correct on 4.28% (55 rows) and only Branch B on 8.25% (106 rows). Branch A's 55 unique wins are led by Zoom_voice (27), GoogleMeet_voice (8) and GoogleMeet_video (7); Branch B's 106 by Discord_video (30), Discord_voice (28) and Zoom_video (27). Zoom_voice recall is 0.712 for Branch A against 0.438 for Branch B; Zoom_video recall is 0.605 against 0.738.
Inference	The errors are partly decorrelated, and Branch A's complementary strength is concentrated in Zoom_voice, a minority class, which is consistent with the recall offsets of Experiment 7.
Verdict	Blending is worth testing; weight selection is done nested.

Experiment C23: Blend weight: full-data sweep and nested grouped selection

Hypothesis	A weighted average $w p_A + (1 - w) p_B$ has an interior optimum, and if the weight is real, the one chosen on the training side of each grouped fold should be stable and its held-out score should hold.
Method	w was swept from 0.00 to 1.00 in steps of 0.05 on all 1,285 rows, with and without the Zoom rule (reference only: weight and score share the same

Table 15. Nested grouped selection of the blend weight (Experiment 23). Pooled: both repeats of every held-out row; fold mean: mean of the ten per-fold scores.

Configuration	Macro-F1 pooled		Macro-F1 fold mean	
	no rule	+ rule	no rule	+ rule
Branch B alone (TabICL, flat+hier)	0.8201	0.8292	0.8175	0.8268
Blend, nested weight, rule-free objective (median 0.40)	0.8319	0.8345	0.8286	0.8318
Blend, nested weight, rule-aware objective (median 0.50)	0.8307	0.8331	—	—
Blend, fixed $w = 0.40$ (optimistic, same rows)	0.8363	0.8366	—	—

rows). Then, for each of the ten grouped folds, w was chosen from the same grid using only that fold’s training rows and applied unchanged to its held-out rows, under two selection objectives (macro-F1 without the rule, and macro-F1 after the rule). Held-out predictions are pooled over both repeats (each row scored twice).

Result	Full-data sweep: without the rule best $w = 0.40$, macro-F1 0.8363 ($w = 0$: 0.8201; $w = 0.5$: 0.8351; $w = 1$: 0.7900); with the rule best $w = 0.50$, 0.8380 ($w = 0.40$: 0.8366; $w = 0$: 0.8292; $w = 1$: 0.7855). Nested selection: Table 15. The rule-free objective selects 0.45, 0.30, 0.40, 0.40, 0.45, 0.40, 0.45, 0.40, 0.25, 0.45 (median 0.40); the rule-aware objective selects 0.55, 0.25, 0.50, 0.55, 0.30, 0.55, 0.50, 0.40, 0.50, 0.50 (median 0.50).
Inference	The optimum is broad (0.40–0.50), the weight is stable across folds (0.25–0.55) and the held-out gain over Branch B alone survives nesting. The full-data figures are optimistic and are used only to locate the region.
Verdict	Weight 0.40/0.60 adopted (the nested median under the rule-free objective and the full-data optimum of the sweep). The headline estimate is the per-fold mean of the nested blend with the rule, 0.8318 . This figure inherits the stratified-fold optimism of Branch A’s probabilities (Section C.11).

Experiment C24: The rule after the blend, and the blend against Branch B

Hypothesis	The rule is orthogonal to which model produced the probabilities, so it should add to the blend as it does to Branch B; and the blend improves on Branch B as shipped (with the rule).
Method	Rule applied to the blend, to Branch B and to Branch A alone, same predictions otherwise. Paired bootstrap over rows (2,000 resamples, both repeats averaged) and McNemar’s test on row correctness, shipped configuration ($w = 0.40$ plus rule) against Branch B plus rule and against raw Branch B.
Result	Blend: 0.8363 $\rightarrow$ 0.8366 (fixed $w = 0.40$) and 0.8319 $\rightarrow$ 0.8345 (nested), i.e. +0.0003 to +0.0026. Branch B alone: 0.8201 $\rightarrow$ 0.8292 (+0.0091). Branch A alone: 0.7900 $\rightarrow$ 0.7855 (−0.0045). Against Branch B plus rule: +0.0072 macro-F1 (pooled 0.8366 against 0.8292; interval [−0.0036, +0.0192], $P(\Delta > 0) = 0.888$); McNemar 16 rows won by the blend against 10 for Branch B, $p = 0.327$. Against raw Branch B: +0.0164 (interval [−0.0007, +0.0330], $P(\Delta > 0) = 0.970$). With the nested weight estimate the gain over Branch B plus rule is +0.0039 (pooled) and +0.0050 (per-fold mean).

Inference	The rule’s value is largest where the model under-predicts Zoom_voice; Branch A already assigns Zoom_voice at high recall, so most of the rule’s effect is absorbed by the blend, and applied to Branch A alone it costs Zoom_video recall. The blend is better in every estimate, by roughly 0.4 to 0.7 macro-F1 points over Branch B with the rule, but the interval includes zero; most of the improvement over <i>raw</i> Branch B is the Zoom rule plus Branch A’s Zoom_voice recall.
Verdict	Blend and rule both retained (threshold 300 bytes, no fitted parameter); the marginal gain of each over the best single branch is reported as not established.

Experiment C25: Final configuration and test-set output

Hypothesis	The validated pipeline can be refit on all training rows and applied to the 327 test flows without change.
Method	Branch A (31 features; the three tree models; $T = 2$; prior correction) was refit on all 1,285 rows, single-threaded; the per-class offsets and the specialist of Experiment 7 were <i>not</i> carried into the blend, so that the blend receives corrected probabilities rather than outputs shifted toward recall. Branch B flat and hierarchical test probabilities (16 estimators, full-data refit) were averaged, with the label order aligned to Branch A’s alphabetical order. Blend $0.4 p_A + 0.6 p_B$, argmax, then the Zoom rule.
Result	The rule routed 43 of 327 test flows to Zoom_voice, in line with the training proportion (an even 80/80 split in the ambiguous subset). Predicted class counts: Discord_voice 67, Discord_video 56, Zoom_voice 43, Messenger_video 33, Zoom_video 27, GoogleMeet_video 26, Messenger_voice 23, WhatsApp_video 22, WhatsApp_voice 19, GoogleMeet_voice 11 (these match the submitted file exactly).
Inference	The output proportions are plausible against the flows-per-call prior of Experiment 1.
Verdict	Submitted. Nested-CV estimate: macro-F1 0.8318.

C.10 Per-Class Results

Table 16 gives the per-class breakdown of the fixed-weight blend.

C.11 Limitations

- **Mixed fold protocols.** Branch A’s out-of-fold probabilities were produced under stratified 5-fold cross-validation, whereas Branch B’s come from the grouped folds, so Branch A’s raw scores in Experiments 22–24 are likely somewhat optimistic relative to a grouped estimate. Correcting this would weaken Branch A, so the selected weight of 0.40 on Branch A is, if anything, too high, and the nested blend estimate of 0.8318 is likely somewhat optimistic.
- **Same-row selection.** The Branch B variant, the 16-estimator setting and the blend were all chosen among near-equal candidates on the same 1,285 rows; every paired difference in Experiments 16 and 24 lies within bootstrap noise.

Table 16. Per-class results of the fixed-weight configuration ($w = 0.40$ plus Zoom rule; pooled, so the mean F1 is 0.8366, not the nested 0.8318) against Branch B plus rule and raw Branch A.

Class	n	Precision	Recall	F1	F1 B+rule / A raw
Discord_video	256	0.969	0.855	0.909	0.905 / 0.846
Discord_voice	238	0.837	0.903	0.869	0.864 / 0.826
GoogleMeet_video	112	0.814	0.741	0.776	0.765 / 0.737
GoogleMeet_voice	40	0.720	0.900	0.800	0.759 / 0.679
Messenger_video	131	0.929	0.893	0.911	0.914 / 0.867
Messenger_voice	94	0.907	0.936	0.921	0.906 / 0.873
WhatsApp_video	82	0.888	0.963	0.924	0.936 / 0.865
WhatsApp_voice	80	0.988	1.000	0.994	0.988 / 0.975
Zoom_video	172	0.978	0.512	0.672	0.672 / 0.703
Zoom_voice	80	0.437	0.912	0.591	0.583 / 0.530

- **Different tuned configurations.** Branch A’s tree settings were tuned for the 31-column set under its own protocol and differ from the Experiment 3 search winners, which were found on the 241-column set.
- **Row-level resampling.** Bootstrap intervals resample rows, not pseudo-groups, and are therefore slightly narrow.
- **Scope of the rule.** The Zoom rule is a property of this dataset’s Zoom audio-only windows (Experiment 18); it is not claimed to transfer to other applications, where voice and video are separable (AUC 0.83–0.88).
- **External and unlabelled data.** Experiment 12 used a public capture corpus (MIRAGE-AppAct-2024) for pretraining and probes only; the final pipeline does not use it. Unlabelled test flows were used for adversarial validation and the domain-adversarial pretraining head, never for fitting a label-dependent quantity.